SET 23 • HSA-REI.B.4.b

FORWARD 1

Find the zeros of a polynomial from its factors • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Find all zeros of $p(x) = (x - 2)(x + 1)(x - 5)$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q1/5

802

QUESTION ID
S23-F1-Q1

TYPE
Short answer

CORRECT ANSWER
2, −1, and 5

GRADING NOTE Form required: all three zeros. Each factor is 0 at one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find all zeros of $p(x) = x^3 - 4x$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q2/5

803

QUESTION ID
S23-F1-Q2

TYPE
Short answer

CORRECT ANSWER
0, 2, and −2

GRADING NOTE Form required: all three zeros. $x^3 - 4x = x(x - 2)(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q2/5

Which polynomial has zeros −3, 0, and 4?

A. x(x + 3)(x − 4)

B. (x + 3)(x − 4)

C. x(x − 3)(x + 4)

D. x(x − 3)(x − 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q3/5

804

QUESTION ID
S23-F1-Q3

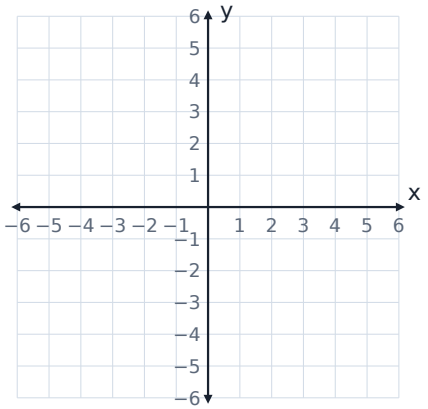
TYPE
Multiple choice

CORRECT ANSWER
A. x(x + 3)(x − 4)

GRADING NOTE A zero at −3 gives the factor x + 3; a zero at 4 gives x − 4; a zero at 0 gives x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q3/5

Sketch a rough graph of $p(x) = -(x + 3)(x - 1)(x - 4)$. Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S23-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q4/5

805

QUESTION ID

S23-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A curve that crosses the x-axis at −3, 1, and 4, passes through (0, −12) (below the grid is fine), rises to the left and falls to the right

GRADING NOTE Any correct sketch: three crossings and the end behavior of a cubic with negative leading coefficient. $p(0) = -(3)(-1)(-4) = -12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S23-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q4/5

True or false?

$p(x) = x^2(x - 3)$ has three different zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q5/5

806

QUESTION ID
S23-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its zeros are 0 (twice) and 3: only two different values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSA-REI.B.4.b

FORWARD 2

Use the Fundamental Theorem of Algebra to count complex zeros • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does a polynomial of degree 5 have?

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q1/5

808

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. By the Fundamental Theorem of Algebra, a degree-n polynomial has n complex zeros counted with multiplicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 - x^2 + 4x - 4 = (x - 1)(x^2 + 4)$.

Find all complex zeros.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q2/5

809

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1, 2i, and −2i

GRADING NOTE Form required: all three zeros. $x^2 + 4 = 0$ gives $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A polynomial with real coefficients has zeros 3 and 2i.

Which number must also be a zero?

A. 3i

B. 2

C. −3

D. −2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q3/5

810

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −2i

GRADING NOTE Nonreal zeros of a polynomial with real coefficients come in conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q3/5

True or false?

$x^2 + 1 = (x + i)(x - i)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q4/5

811

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + i)(x - i) = x^2 - i^2 = x^2 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q4/5

True or false?

Every polynomial of degree 3 has three real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q5/5

812

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^3 + x = x(x^2 + 1)$ has only one real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q5/5

Solve quadratic equations with the quadratic formula and use the discriminant

MAIN	Algebra I main question	**HSA-REI.B.4.b**
BACKWARD 1	Complete the square to solve a quadratic equation	**HSA-REI.B.4.a**
BACKWARD 2	Estimate square roots with rational approximations	**8.NS.A.2**
FORWARD 1	Solve right-triangle problems that lead to quadratic equations	**HSG-SRT.C.8**
FORWARD 2	Solve quadratic equations with complex solutions	**HSN-CN.C.7**

SET 24 • HSA-REI.B.4.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $2x^2 - 3x - 4 = 0$ with the quadratic formula.

Give exact answers in simplest radical form.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S24-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q1/5

815

QUESTION ID

S24-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x = (3 + \sqrt{41}) \div 4$ or
$x = (3 - \sqrt{41}) \div 4$

GRADING NOTE REQUIRED WORK: a = 2, b = −3, and c = −4 substituted into $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$, with the discriminant 9 + 32 = 41 shown.

Form required: exact, in simplest radical form (both solutions), with the formula work shown. Decimals are not exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S24-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q1/5

Solve $x^2 + 4x - 1 = 0$. Round each solution to the nearest hundredth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q2/5

816

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x \approx 0.24$ or $x \approx -4.24$

GRADING NOTE Form required: both solutions, rounded to the nearest hundredth. $x = -2 \pm \sqrt{5}$, and $\sqrt{5} \approx 2.236$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q2/5

Use the discriminant to find how many real solutions $3x^2 - 2x + 5 = 0$ has.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q3/5

817

QUESTION ID

S24-M-Q3

TYPE

Short answer

CORRECT ANSWER

None. The discriminant is −56.

GRADING NOTE Form required: the number of real solutions and the discriminant. $b^2 - 4ac = 4 - 60 = -56 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q3/5

Solve $x^2 + 2x + 7 = 0$ over the real numbers.

A. No real solution

B. $x = -1 + \sqrt{6}$ or $x = -1 - \sqrt{6}$

C. $x = -7$ or $x = 1$

D. $x = 1 + \sqrt{6}$ or $x = 1 - \sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q4/5

818

QUESTION ID

S24-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. No real solution

GRADING NOTE $b^2 - 4ac = 4 - 28 = -24 < 0$, so no real number is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q4/5

True or false?

The equation $4x^2 - 12x + 9 = 0$ has exactly one real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q5/5

819

QUESTION ID
S24-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $b^2 - 4ac = 144 - 144 = 0$. The equation is $(2x - 3)^2 = 0$, so $x = 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q5/5

SET 24 • HSA-REI.B.4.b

BACKWARD 1

Complete the square to solve a quadratic equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 4x = 5$ by completing the square.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q1/5

821

QUESTION ID

S24-B1-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x = 5$ or $x = -1$

GRADING NOTE REQUIRED WORK: 4 added to both sides to get $(x - 2)^2 = 9$, then $x - 2 = 3$ or $x - 2 = -3$.

Form required: both solutions, with the completed square shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x - 2 = 0$ by completing the square. Give exact answers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q2/5

822

QUESTION ID

S24-B1-Q2

TYPE

Answer with required work

CORRECT ANSWER

$x = -1 + \sqrt{3}$ or $x = -1 - \sqrt{3}$

GRADING NOTE REQUIRED WORK: $x^2 + 2x = 2$, then 1 added to both sides to get $(x + 1)^2 = 3$, then $x + 1 = \pm\sqrt{3}$.

Form required: exact (both solutions), with the completed square shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation results from completing the square on $x^2 - 6x + 4 = 0$?

A. $(x + 3)^2 = 5$

B. $(x - 3)^2 = 5$

C. $(x - 3)^2 = 13$

D. $(x - 6)^2 = 32$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q3/5

823

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x - 3)^2 = 5$

GRADING NOTE $x^2 - 6x = -4$; add 9 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q3/5

True or false?

Adding 25 to both sides of $x^2 + 10x = 3$ gives $(x + 5)^2 = 28$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q4/5

824

QUESTION ID

S24-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 + 10x + 25 = (x + 5)^2$, and $3 + 25 = 28$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q4/5

True or false?

Completing the square on $x^2 + 6x = 1$ gives $(x + 3)^2 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q5/5

825

QUESTION ID
S24-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Add 9 to both sides: $(x + 3)^2 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSA-REI.B.4.b

BACKWARD 2

Estimate square roots with rational approximations • Grade 8 • The Number System • 8.NS.A.2

Between which two consecutive whole numbers is √41?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q1/5

827

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 and 7

GRADING NOTE Any equivalent form. 36 < 41 < 49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round √5 to the nearest hundredth.

√5 ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q2/5

828

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2.24

GRADING NOTE Form required: rounded to the nearest hundredth. √5 ≈ 2.236.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q2/5

Which is the best estimate of √30?

A. 6.2

B. 5.0

C. 15

D. 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q3/5

829

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5.5

GRADING NOTE √30 ≈ 5.48: $5.5^2 = 30.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{2} \approx 1.41$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q4/5

830

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1.41^2 = 1.9881$, close to 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

√90 is between 8 and 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q5/5

831

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 81 < 90 < 100, so √90 is between 9 and 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSA-REI.B.4.b

FORWARD 1

Solve right-triangle problems that lead to quadratic equations • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle's legs are x and x + 7, and its hypotenuse is 13.

Write an equation and find the lengths of the legs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q1/5

833

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + (x + 7)^2 = 169$; the legs are 5 and 12.

GRADING NOTE Form required: an equation and both legs. $x^2 + 7x - 60 = 0$, so $x = 5$ or $x = -12$; a length cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q1/5

A 10-ft ladder leans against a wall and makes a 70° angle with the ground.

How high up the wall does it reach, to the nearest tenth of a foot?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q2/5

834

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9.4 feet

GRADING NOTE Form required: rounded to the nearest tenth. h = 10 sin 70° ≈ 9.40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q2/5

A rectangle's diagonal is 10, and its length is 2 more than its width.

What is the width?

A. −8

B. 6

C. 8

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q3/5

835

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE $w^2 + (w + 2)^2 = 100$ gives $w^2 + 2w - 48 = 0$, so $w = 6$ or $w = -8$. A width must be positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q3/5

True or false?

A ramp rises 1 ft over a horizontal run of 12 ft. Its angle with the ground is about 4.8°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q4/5

836

QUESTION ID
S24-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $\tan\theta = 1 \div 12$, so $\theta = \tan^{-1}(1/12) \approx 4.76°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs 1 and 1 has a hypotenuse of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q5/5

837

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hypotenuse is √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSA-REI.B.4.b

FORWARD 2

Solve quadratic equations with complex solutions • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 - 4x + 13 = 0$. Write the solutions in the form $a + bi$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q1/5

839

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 2 + 3i or

x = 2 − 3i

GRADING NOTE Form required: a + bi (both solutions). $x = (4 \pm \sqrt{(-36)}) \div 2 = 2 \pm 3i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q1/5

Solve $x^2 + 25 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q2/5

840

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 5i$ or $x = -5i$

GRADING NOTE Form required: both solutions. $x^2 = -25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q2/5

What are the solutions of $x^2 + 2x + 5 = 0$?

A. $1 + 2i$ and $1 - 2i$

B. $-1 + 4i$ and $-1 - 4i$

C. $-2 + 4i$ and $-2 - 4i$

D. $-1 + 2i$ and $-1 - 2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q3/5

841

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −1 + 2i and −1 − 2i

GRADING NOTE $x = (-2 \pm \sqrt{(4 - 20)}) \div 2 = (-2 \pm 4i) \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q3/5

True or false?

If a quadratic equation with real coefficients has a negative discriminant, its two solutions are complex conjugates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q4/5

842

QUESTION ID
S24-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The ± in the quadratic formula gives a + bi and a − bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q4/5

True or false?

The solutions of $x^2 + 2x + 2 = 0$ are $1 + i$ and $1 - i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q5/5

843

QUESTION ID
S24-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x = (-2 \pm \sqrt{(-4)}) \div 2 = -1 \pm i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q5/5

Create quadratic and exponential equations in one variable and use them to solve problems

MAIN	Algebra I main question	**HSA-CED.A.1**
BACKWARD 1	Solve quadratic equations	**HSA-REI.B.4.b**
BACKWARD 2	Use properties of integer exponents	**8.EE.A.1**
FORWARD 1	Use volume formulas to write and solve equations for a dimension	**HSG-GMD.A.3**
FORWARD 2	Solve exponential models with logarithms	**HSF-LE.A.4**

SET 25 • HSA-CED.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A ball's height is $h = -16t^2 + 48t + 4$ feet after t seconds.

Write and solve an equation to find when the ball is 36 feet high.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q1/5

846

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-16t^2 + 48t + 4 = 36$; t = 1 second and t = 2 seconds

GRADING NOTE Form required: an equation and both times. $-16t^2 + 48t - 32 = 0$, so $t^2 - 3t + 2 = 0$ and $(t - 1)(t - 2) = 0$. The ball passes 36 ft going up and coming down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle's length is 5 cm more than its width. Its area is 84 cm².

Write an equation and find the dimensions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q2/5

847

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

w(w + 5) = 84; 7 cm by 12 cm

GRADING NOTE Form required: an equation and both dimensions. $w^2 + 5w - 84 = (w + 12)(w - 7) = 0$; a width cannot be −12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q2/5

A culture of 200 bacteria doubles every hour.

Write an equation for the number of hours t until there are 6,400 bacteria, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q3/5

848

QUESTION ID

S25-M-Q3

TYPE

Short answer

CORRECT ANSWER

$200 \cdot 2^t = 6{,}400$;

t = 5 hours

GRADING NOTE Form required: an equation and its solution. $2^t = 32 = 2^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q3/5

A $2,000 investment earns 5% interest, added once at the end of each year.

Which inequality gives the whole numbers of years n after which it is worth at least $3,000?

A. $2{,}000(1.5)^n \geq 3{,}000$

B. $2{,}000(0.05)^n \geq 3{,}000$

C. $2{,}000 + 0.05n \geq 3{,}000$

D. $2{,}000(1.05)^n \geq 3{,}000$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q4/5

849

QUESTION ID

S25-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D.

$2{,}000(1.05)^n \geq 3{,}000$

GRADING NOTE Each year the value is multiplied by 1 + 0.05 = 1.05. The first whole year it reaches $3,000 is n = 9: $2{,}000(1.05)^8 \approx \$2{,}954.91$ and $2{,}000(1.05)^9 \approx \$3{,}102.66$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q4/5

True or false?

A square garden of side x ft has a 2-ft path around it, and the garden and path cover 144 ft². Then $(x + 4)^2 = 144$, so $x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q5/5

850

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The path adds 2 ft on each side, so the outer side is x + 4 = 12. The other root, x = −16, is not a length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q5/5

SET 25 • HSA-CED.A.1

BACKWARD 1

Solve quadratic equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $t^2 - 3t + 2 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q1/5

852

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

t = 1 or t = 2

GRADING NOTE Form required: both solutions. $(t - 1)(t - 2) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $w^2 + 5w = 84$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q2/5

853

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

w = 7 or w = −12

GRADING NOTE Form required: both solutions. (w + 12)(w − 7) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q2/5

Solve $(x + 4)^2 = 144$.

A. $x = 12$ or $x = -12$

B. $x = 140$ or $x = -148$

C. $x = 8$ only

D. $x = 8$ or $x = -16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q3/5

854

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = 8$ or $x = -16$

GRADING NOTE $x + 4 = 12$ or $x + 4 = -12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q3/5

True or false?

$x = -2$ is a solution of $x^2 + 5x + 6 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q4/5

855

QUESTION ID
S25-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $4 - 10 + 6 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q4/5

True or false?

The only solution of $x^2 = 3x$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q5/5

856

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x - 3) = 0$, so $x = 0$ is also a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSA-CED.A.1

BACKWARD 2

Use properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 32 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q1/5

858

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2^5

GRADING NOTE Form required: a power of 2. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $2^3 \cdot 2^4$. Write the answer as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q2/5

859

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2^7

GRADING NOTE Form required: a power of 2. Add the exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which power of 2 equals 64?

A. 2^{32}

B. 2^{8}

C. 2^{5}

D. 2^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q3/5

860

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2^{6}

GRADING NOTE $2^{6} = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q3/5

True or false?

$3^4 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q4/5

861

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 \cdot 3 \cdot 3 \cdot 3 = 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q4/5

True or false?

$2^{-3} = \frac{1}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q5/5

862

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2^{-3} = 1 \div 2^3 = 1 \div 8$. A negative exponent does not make the value negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSA-CED.A.1

FORWARD 1

Use volume formulas to write and solve equations for a dimension • High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has height 10 cm and volume 250π cm^3.

Write an equation and find its radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q1/5

864

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\pi r^2(10) = 250\pi$; r = 5 cm

GRADING NOTE Form required: an equation and the radius. $r^2 = 25$, and a radius is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q1/5

A square pyramid has height 9 in and volume 300 in³.

Find the side length of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q2/5

865

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 inches

GRADING NOTE Any equivalent form. $\frac{1}{3}s^2(9) = 300$, so $3s^2 = 300$ and $s^2 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q2/5

A cone has volume 12π and height 4.

What is its radius?

A. 9

B. 6

C. $\sqrt{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q3/5

866

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE $\frac{1}{3}\pi r^2(4) = 12\pi$, so $r^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q3/5

True or false?

A sphere of radius 3 has volume 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q4/5

867

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\frac{4}{3}\pi(27) = 36\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q4/5

True or false?

Doubling the radius of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q5/5

868

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The volume depends on r^2, so it is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSA-CED.A.1

FORWARD 2

Solve exponential models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $200 \cdot 2^t = 5{,}000$. Give the exact answer using logarithms, then round it to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q1/5

870

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$t = \log 25 \div \log 2 \approx 4.64$

GRADING NOTE Form required: the logarithm expression and the rounded value. $2^t = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q1/5

Solve $2{,}000(1.05)^n = 3{,}000$. Round to the nearest tenth.

n ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q2/5

871

QUESTION ID
S25-F2-Q2

TYPE
Short answer

CORRECT ANSWER
8.3

GRADING NOTE Form required: rounded to the nearest tenth. n = log 1.5 ÷ log 1.05 ≈ 8.31.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q2/5

Solve $10^x = 500$. (log means base 10.)

A. x = 2.5

B. x = 5,000

C. x = 50

D. x = log 500 ≈ 2.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q3/5

872

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = log 500 ≈ 2.70

GRADING NOTE x = log 500 because log 500 is the power of 10 that gives 500; log 500 ≈ 2.699.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q3/5

True or false?

The solution of $e^x = 7$ is $x = \ln 7$. (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q4/5

873

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ln is the logarithm base e.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q4/5

True or false?

The solution of $5^x = 30$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q5/5

874

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5^x = 30$ gives $x = \log 30 \div \log 5 \approx 2.11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q5/5

Show that replacing an equation by its sum with a multiple of the other keeps the solutions of a system

MAIN	Algebra I main question	**HSA-REI.C.5**
BACKWARD 1	Justify steps with the properties of equality	**HSA-REI.A.1**
BACKWARD 2	Recognize the solution of a system as a point on both lines	**8.EE.C.8.a**
FORWARD 1	Write a system as a matrix equation AX = B and use row operatio..	**HSA-REI.C.8**
FORWARD 2	Use determinants and inverse matrices	**HSN-VM.C.10**

SET 26 • HSA-REI.C.5

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

ANSWER KEY Reduced copy of the question page, with its answer

System: x + y = 7 and x − y = 1.

Replace the second equation by the sum of the two equations. Write the new system, solve it, and show that the solution also satisfies the original system.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q1/5

877

QUESTION ID

S26-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

(4, 3)

GRADING NOTE REQUIRED WORK: The new system x + y = 7 and 2x = 8, its solution, and a check of the solution in both original equations.

Form required: the new system, the solution, and the check. 2x = 8 gives x = 4, so y = 3. Check: 4 + 3 = 7 and 4 − 3 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Suppose (a, b) is a solution of the system $2x + y = 5$ and $x - 3y = 6$.

Explain why (a, b) is also a solution of $(2x + y) + 2(x - 3y) = 5 + 2(6)$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q2/5

878

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

Since 2a + b = 5 and a − 3b = 6, 2(a − 3b) = 12. Adding equal amounts to both sides of 2a + b = 5 gives (2a + b) + 2(a − 3b) = 5 + 12, a true equation.

GRADING NOTE Any equivalent explanation that substitutes (a, b) and adds equal quantities to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q2/5

System: 3x + 2y = 12 and x − y = −1.

Which system has the same solution?

A. 3x + 2y = 12 and 4x + y = 13

B. 3x + 2y = 12 and 2x + 3y = 11

C. 3x + 2y = 12 and 5x = 14

D. 3x + 2y = 12 and 5x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q3/5

879

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3x + 2y = 12 and 5x = 10

GRADING NOTE Adding the first equation and 2 times the second gives 5x = 10. The solution of both systems is (2, 3); (2, 3) does not satisfy 5x = 14, 4x + y = 13, or 2x + 3y = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q3/5

True or false?

Replacing one equation of a system by the sum of that equation and 3 times the other equation can change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q4/5

880

QUESTION ID
S26-M-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Any solution of the old system satisfies the new equation, and the old equation can be recovered by subtracting 3 times the other, so the solutions stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q4/5

True or false?

In the system 2x + 3y = 7 and 4x − 3y = 5, adding the equations eliminates y and gives 6x = 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q5/5

881

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3y + (−3y) = 0 and 7 + 5 = 12. Then x = 2 and y = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q5/5

SET 26 • HSA-REI.C.5

BACKWARD 1

Justify steps with the properties of equality • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If a = b and c = d, explain why a + c = b + d.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q1/5

883

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Adding c to both sides of a = b gives a + c = b + c. Since c = d, replace c by d on the right: a + c = b + d.

GRADING NOTE Any equivalent explanation using the addition property and substitution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q1/5

Name the property used to go from x = 5 to 2x = 10.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q2/5

884

QUESTION ID
S26-B1-Q2

TYPE
Short answer

CORRECT ANSWER
Multiplication property of equality (multiply both sides by 2)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If p = q, which equation must also be true?

A. 4p = q

B. p = q + 4

C. p + 4 = q + 4

D. p + 4 = q

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q3/5

885

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. p + 4 = q + 4

GRADING NOTE Add 4 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If a = b, then 5a = 5b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q4/5

886

QUESTION ID

S26-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiplication property of equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If a = b and c = d, then a − c = b + d.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q5/5

887

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtracting equal amounts gives a − c = b − d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSA-REI.C.5

BACKWARD 2

Recognize the solution of a system as a point on both lines • Grade 8 • Expressions & Equations • 8.EE.C.8.a

Is (2, 5) a solution of the system y = 2x + 1 and y = x + 3? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q1/5

889

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(2) + 1 = 5 and 2 + 3 = 5.

GRADING NOTE Any equivalent explanation that checks both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The lines y = x + 1 and y = −x + 5 meet at (2, 3).

What is the solution of the system?

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q2/5

890

QUESTION ID

S26-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 3)

GRADING NOTE Any equivalent form, such as x = 2, y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q2/5

Which point is a solution of both $x + y = 6$ and $x - y = 2$?

A. (2, 4)

B. (4, 2)

C. (3, 3)

D. (5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q3/5

891

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (4, 2)

GRADING NOTE $4 + 2 = 6$ and $4 - 2 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q3/5

True or false?

The solution of a system of two linear equations is the point where their graphs intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q4/5

892

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That point lies on both lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q4/5

True or false?

(1, 1) is a solution of the system y = 2x and y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q5/5

893

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2(1) = 2, not 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSA-REI.C.5

FORWARD 1

Write a system as a matrix equation AX = B and use row operations • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

Write the system 2x + y = 7 and x − 3y = 0 as a matrix equation AX = B. Give A, X, and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q1/5

895

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A has rows [2 1] and [1 −3]; X is the column [x, y]; B is the column [7, 0]; and AX = B.

GRADING NOTE Form required: A, X, and B, in this order of rows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the matrix equation AX = B, A has rows [1 2] and [3 −1], X is the column [x, y], and B is the column [5, 1].

Write the system it represents.

System: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q2/5

896

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x + 2y = 5 and 3x − y = 1

GRADING NOTE Any equivalent form. Each row of A times X equals the matching entry of B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q2/5

On the rows [1 2 | 5] and [3 −1 | 1], replace row 2 by (row 2) − 3 × (row 1).

What is the new row 2?

A. [0 −7 | 14]

B. [2 −3 | −4]

C. [0 −7 | −14]

D. [0 5 | 16]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q3/5

897

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. [0 −7 | −14]

GRADING NOTE 3 − 3 = 0, −1 − 6 = −7, and 1 − 15 = −14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q3/5

True or false?

Adding a multiple of one row of an augmented matrix to another row does the same thing as replacing that equation by its sum with a multiple of the other equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q4/5

898

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each row is an equation, so this row operation is the elimination step written in matrix form. (Swapping rows or multiplying a row by a nonzero number are other row operations.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q4/5

True or false?

The coefficient matrix of x + y = 3 and 2x − y = 0 has the rows [1 1] and [2 1].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q5/5

899

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second row is [2 −1].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-REI.C.5

FORWARD 2

Use determinants and inverse matrices • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10

Find the determinant of the matrix with rows [3 1] and [5 2].

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q1/5

901

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. 3(2) − 1(5) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q1/5

Show that the matrix with rows [2 −1] and [−5 3] is the inverse of the matrix with rows [3 1] and [5 2].

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q2/5

902

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The product has rows [3(2) + 1(−5) 3(−1) + 1(3)] = [1 0] and [5(2) + 2(−5) 5(−1) + 2(3)] = [0 1], the identity matrix.

GRADING NOTE Any equivalent work that multiplies the matrices and gets the identity matrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q2/5

Which matrix has no inverse?

A. Rows [2 4] and [1 2]

B. Rows [1 0] and [0 1]

C. Rows [2 1] and [1 1]

D. Rows [3 1] and [5 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q3/5

903

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Rows [2 4] and [1 2]

GRADING NOTE 2(2) − 4(1) = 0. A 2-by-2 matrix has an inverse only when its determinant is not 0; each of the other three matrices has determinant 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system of two linear equations has an inverse, the system has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q4/5

904

QUESTION ID
S26-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The solution is $X = A^{-1}B$, and it is the only one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q4/5

True or false?

The inverse of the matrix with rows [2 0] and [0 4] has rows [−2 0] and [0 −4].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q5/5

905

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The inverse has rows $[\frac{1}{2}\ 0]$ and $[0\ \frac{1}{4}]$: each diagonal entry is replaced by its reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q5/5

Prove that replacing an equation by its sum with a multiple of the other gives an equivalent system, in both directions

MAIN	Algebra I main question	**HSA-REI.C.5**
BACKWARD 1	Use the properties of equality to justify a step	**HSA-REI.A.1**
BACKWARD 2	Solve simple systems by adding equations	**8.EE.C.8.b**
FORWARD 1	Use row operations on an augmented matrix as elimination steps	**HSA-REI.C.8**
FORWARD 2	Solve the same system with an inverse matrix	**HSA-REI.C.9**

SET 72 • HSA-REI.C.5

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

ANSWER KEY Reduced copy of the question page, with its answer

System A: $x + 2y = 7$ and $3x - y = 7$. System B: $x + 2y = 7$ and $7x = 21$, where $7x = 21$ is the first equation plus 2 times the second.

Show that every solution of A solves B and every solution of B solves A. Then give the solution.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q1/5

908

QUESTION ID

S72-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

(3, 2)

GRADING NOTE REQUIRED WORK: A to B: a solution of A makes both equations true, so adding $x + 2y = 7$ and $2(3x - y) = 14$ gives $7x = 21$. B to A: from $7x = 21$ and $x + 2y = 7$, subtracting gives $6x - 2y = 14$, so $3x - y = 7$. The kept equation $x + 2y = 7$ is in both systems.

Form required: both directions and the solution. Check: $3 + 4 = 7$ and $9 - 2 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q1/5

Let P = c and Q = d be the two equations of a system (P and Q are expressions in x and y). Prove that the system P = c and Q + kP = d + kc has exactly the same solutions, for any number k.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q2/5

909

QUESTION ID

S72-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

The two systems have the same solutions.

GRADING NOTE REQUIRED WORK: If (x, y) makes P = c and Q = d true, then Q + kP = d + kc (adding equal amounts), so it solves the new system. Conversely, if P = c and Q + kP = d + kc, subtracting kP = kc gives Q = d. So the solution sets are equal.

Form required: both directions, each with its reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q2/5

Why must the new system keep one of the original equations unchanged? Give an example.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q3/5

910

QUESTION ID

S72-M-Q3

TYPE

Short answer

CORRECT ANSWER

The kept equation is needed to recover the replaced one. For example, replacing both x + y = 2 and x − y = 0 with their sum 2x = 2 alone loses information: 2x = 2 has infinitely many solutions (x = 1, any y), but the system has only (1, 1).

GRADING NOTE Any equivalent explanation with an example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q3/5

System: 2x + y = 5 and x − y = 1. Which step can produce a system with DIFFERENT solutions?

A. Replacing both equations by their sum, 3x = 6, alone

B. Multiplying the first equation by 3

C. Writing the equations in the other order

D. Replacing the second equation by the sum of the two equations

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q4/5

911

QUESTION ID
S72-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. Replacing both equations by their sum, 3x = 6, alone

GRADING NOTE 3x = 6 alone allows any y. The other steps keep the solution (2, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q4/5

True or false?

If (2, 1) solves x + y = 3 and x − y = 1, then (2, 1) also solves (x + y) + 4(x − y) = 3 + 4(1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q5/5

912

QUESTION ID
S72-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 3 + 4(1) = 7 on both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q5/5

SET 72 • HSA-REI.C.5

BACKWARD 1

Use the properties of equality to justify a step • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If P = c and Q = d are true, explain why P + Q = c + d is true.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q1/5

914

QUESTION ID

S72-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Add Q to both sides of P = c to get P + Q = c + Q; since Q = d, replace Q by d on the right.

GRADING NOTE Any equivalent explanation using the addition property and substitution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q1/5

If P + 3Q = c + 3d and Q = d, explain why P = c.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q2/5

915

QUESTION ID

S72-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Since Q = d, 3Q = 3d. Subtracting equal amounts from both sides of P + 3Q = c + 3d gives P = c.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q2/5

Which property justifies going from $x - y = 1$ to $4x - 4y = 4$?

A. Addition property of equality

B. Commutative property

C. Substitution of y

D. Multiplication property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q3/5

916

QUESTION ID

S72-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Multiplication property of equality

GRADING NOTE Both sides are multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q3/5

True or false?

Subtracting equal quantities from both sides of a true equation gives a true equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q4/5

917

QUESTION ID
S72-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Subtraction property of equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q4/5

True or false?

If a + b = 10 and a = 4, then b = 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q5/5

QUESTION ID

S72-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE b = 10 − 4 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSA-REI.C.5

BACKWARD 2

Solve simple systems by adding equations • Grade 8 • Expressions & Equations • 8.EE.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve by adding the equations.

$x + y = 9$

$x - y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q1/5

920

QUESTION ID

S72-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(6, 3)

GRADING NOTE Any equivalent form. 2x = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$2x + y = 11$

$y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q2/5

921

QUESTION ID

S72-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 3)

GRADING NOTE Any equivalent form. 2x = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q2/5

Solve.

$x + 2y = 8$

$x = 2$

A. (2, 5)

B. (8, 0)

C. (2, 3)

D. (3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q3/5

922

QUESTION ID

S72-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 3)

GRADING NOTE 2 + 2y = 8, so y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Adding $x + y = 5$ and $x - y = 1$ gives $2x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q4/5

923

QUESTION ID

S72-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE y and −y cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q4/5

True or false?

The system x + y = 4 and 2x + 2y = 8 has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q5/5

924

QUESTION ID

S72-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second equation is 2 times the first: infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSA-REI.C.5

FORWARD 1

Use row operations on an augmented matrix as elimination steps • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Write the augmented matrix of the system x + 2y = 7 and 3x − y = 7.

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q1/5

926

QUESTION ID

S72-F1-Q1

TYPE

Short answer

CORRECT ANSWER

[1 2 | 7] and [3 −1 | 7]

GRADING NOTE Form required: both rows in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q1/5

Apply R2 → 2R2 + R1 to the rows [1 2 | 7] and [3 −1 | 7]. What is the new second row, and which equation does it represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q2/5

927

QUESTION ID
S72-F1-Q2

TYPE
Short answer

CORRECT ANSWER
[7 0 | 21], which is 7x = 21

GRADING NOTE Form required: the row and its equation. 2[3 −1 | 7] + [1 2 | 7].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q2/5

Which row operation matches "replace equation 2 by itself minus 3 times equation 1"?

A. R2 → R1

B. R2 → R2 − 3R1

C. R2 → 3R2 − R1

D. R1 → R1 − 3R2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q3/5

928

QUESTION ID

S72-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. R2 → R2 − 3R1

GRADING NOTE Row 2 changes; row 1 is kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q3/5

True or false?

A row operation R2 → R2 + kR1 does not change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q4/5

929

QUESTION ID
S72-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE It is the elimination step proved in the MAIN section.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q4/5

True or false?

Replacing R2 by 0 · R2 + R1 keeps the solutions of every system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q5/5

930

QUESTION ID

S72-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It discards equation 2, so the solution set can grow.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSA-REI.C.5

FORWARD 2

Solve the same system with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

ANSWER KEY Reduced copy of the question page, with its answer

For x + 2y = 7 and 3x − y = 7, the coefficient matrix A has rows [1 2] and [3 −1]. Find the determinant of A.

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q1/5

932

QUESTION ID

S72-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−7

GRADING NOTE Any equivalent form. 1(−1) − 2(3) = −7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A^{-1} is $(1 \div (-7))$ times the matrix with rows $[-1\ -2]$ and $[-3\ 1]$. Use $X = A^{-1}B$, with B the column $[7, 7]$, to solve the system.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q2/5

933

QUESTION ID

S72-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3, y = 2

GRADING NOTE Form required: both values. $(1 \div (-7))$ times the column $[-7 - 14, -21 + 7] = [3, 2]$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q2**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q2/5

If the determinant of the coefficient matrix of a 2-by-2 linear system is 0, which is true?

A. $X = A^{-1}B$ gives the solution.

B. The system has exactly one solution.

C. The system has no solution or infinitely many solutions.

D. The two equations must be identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q3/5

934

QUESTION ID

S72-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The system has no solution or infinitely many solutions.

GRADING NOTE A has no inverse, so the system does not have exactly one solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q3/5

True or false?

Solving $AX = B$ with $X = A^{-1}B$ gives the same solution as elimination.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q4/5

935

QUESTION ID

S72-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both methods find the one solution when A has an inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q4/5

True or false?

For all 2-by-2 matrices A and B, $A^{-1}B = BA^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q5/5

936

QUESTION ID

S72-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Matrix multiplication is not commutative in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q5/5

Solve systems of linear equations exactly by substitution and elimination

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Eliminate a variable by adding a multiple of an equation	**HSA-REI.C.5**
BACKWARD 2	Solve simple systems of linear equations algebraically	**8.EE.C.8.b**
FORWARD 1	Use slopes and systems to find where parallel or perpendicular li..	**HSG-GPE.B.5**
FORWARD 2	Multiply a matrix by a column to check a solution	**HSN-VM.C.8**

SET 27 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Solve by substitution.

$y = 3x - 5$

$2x + y = 15$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q1/5

939

QUESTION ID

S27-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

(4, 7)

GRADING NOTE REQUIRED WORK: 3x − 5 substituted for y: 2x + (3x − 5) = 15 solved for x, then y found from y = 3x − 5.

Form required: the solution with the substitution shown; it may be written x = 4, y = 7. Check: 8 + 7 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q1/5

Solve by elimination.

$3x + 4y = 10$

$5x - 2y = 34$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q2/5

940

QUESTION ID

S27-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

(6, −2)

GRADING NOTE

REQUIRED WORK: The second equation multiplied by 2 and added to the first, so y cancels (13x = 78), then y found by substitution.

Form required: the solution with the elimination shown; it may be written x = 6, y = −2. Check: 30 + 4 = 34.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q2/5

Solve.

$\frac{1}{2}x + \frac{1}{3}y = 4$

$x - y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q3/5

941

QUESTION ID

S27-M-Q3

TYPE

Short answer

CORRECT ANSWER

(6, 3)

GRADING NOTE Any equivalent form. Multiply the first equation by 6: 3x + 2y = 24. Substitute x = y + 3: 5y + 9 = 24, so y = 3. Check: 3 + 1 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q3/5

Solve.

$2x - 3y = -1$

$4x + y = 19$

A. $(-4, -3)$

B. $(4, 3)$

C. $(3, 4)$

D. $(5, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q4/5

942

QUESTION ID

S27-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (4, 3)

GRADING NOTE $y = 19 - 4x$, so $2x - 57 + 12x = -1$ and $x = 4$, $y = 3$. Check: $8 - 9 = -1$ and $16 + 3 = 19$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q4/5

True or false?

The system 2x + y = 5 and 4x + 2y = 9 has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q5/5

943

QUESTION ID

S27-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Doubling the first equation gives 4x + 2y = 10, which cannot also equal 9. The lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 1

Eliminate a variable by adding a multiple of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Add the equations x + 2y = 9 and 3x − 2y = 3.

What equation results, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q1/5

945

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4x = 12; x = 3

GRADING NOTE Form required: the new equation and x. The y-terms cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q1/5

System: 2x + y = 8 and 6x − 5y = 8.

What multiple of the first equation should you subtract from the second to eliminate x?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q2/5

946

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 6x − 3(2x) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q2/5

System: x + y = 5 and 2x + 3y = 12.

Which equation can replace the second equation without changing the solution?

A. x = 2

B. y = 7

C. y = 2

D. 3x + 4y = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q3/5

947

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2

GRADING NOTE (2x + 3y) − 2(x + y) = 12 − 10 gives y = 2. The solution is (3, 2), which does not satisfy x = 2, 3x + 4y = 12, or y = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q3/5

True or false?

Multiplying one equation of a system by a nonzero number does not change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q4/5

948

QUESTION ID
S27-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The new equation has exactly the same solutions as the old one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q4/5

True or false?

Adding the equations $x + y = 4$ and $x - y = 2$ gives $2x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q5/5

949

QUESTION ID
S27-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $4 + 2 = 6$, so $2x = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 2

Solve simple systems of linear equations algebraically • Grade 8 • Expressions & Equations • 8.EE.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$y = x + 2$

$y = 3x$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q1/5

951

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(1, 3)

GRADING NOTE Any equivalent form. $3x = x + 2$, so $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

x + y = 10

y = 4

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q2/5

952

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(6, 4)

GRADING NOTE Any equivalent form. x + 4 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q2/5

Solve.

$y = 2x$

$x + y = 12$

A. (4, 8)

B. (6, 6)

C. (8, 4)

D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q3/5

953

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (4, 8)

GRADING NOTE $x + 2x = 12$, so $x = 4$ and $y = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q3/5

True or false?

The system $y = 2x + 3$ and $y = 2x - 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q4/5

954

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same slope, different y-intercepts: the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q4/5

True or false?

The solution of $y = x$ and $y = -x + 6$ is $(2, 4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q5/5

955

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = -x + 6$ gives $x = 3$, so the solution is $(3, 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 1

Use slopes and systems to find where parallel or perpendicular lines meet • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is $y = 2x + 1$. Write an equation of the line through $(5, 1)$ that is perpendicular to ℓ, and find where the two lines meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q1/5

957

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{1}{2}x + 3.5$; the lines meet at (1, 3).

GRADING NOTE Any equivalent equation; the point is required. The slope is $-\frac{1}{2}$. $2x + 1 = -\frac{1}{2}x + 3.5$ gives $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are the lines $2x + 3y = 6$ and $3x - 2y = 4$ perpendicular? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q2/5

958

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Their slopes are $-\frac{2}{3}$ and $\frac{3}{2}$, and the product is -1.

GRADING NOTE Any equivalent explanation using slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which line is parallel to $y = 3x - 2$ and passes through $(0, 5)$?

A. $y = 5x + 3$

B. $y = -3x + 5$

C. $y = 3x + 5$

D. $y = -\frac{1}{3}x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q3/5

959

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 3x + 5

GRADING NOTE Parallel lines have the same slope, 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q3/5

True or false?

The lines $y = 4x + 1$ and $y = 4x - 7$ have no point in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q4/5

960

QUESTION ID
S27-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE They are parallel and distinct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q4/5

True or false?

The lines y = 2x and y = −2x are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q5/5

961

QUESTION ID
S27-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 2(−2) = −4, not −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 2

Multiply a matrix by a column to check a solution • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8

Multiply the matrix with rows [2 1] and [1 −3] by the column [4, 1].

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q1/5

963

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The column [9, 1]

GRADING NOTE Any equivalent form. 2(4) + 1(1) = 9 and 1(4) − 3(1) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q1/5

The matrix A has rows [2 1] and [1 −3].

Use multiplication by A to decide whether x = 4, y = 1 solves 2x + y = 9 and x − 3y = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q2/5

964

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. A times the column [4, 1] is the column [9, 1], which matches the constants.

GRADING NOTE Any equivalent explanation using the product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q2/5

What is the product of the matrix with rows [1 2] and [3 4] and the column [1, 1]?

A. The column [3, 7]

B. The column [2, 6]

C. The column [10]

D. The column [4, 6]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q3/5

965

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The column [3, 7]

GRADING NOTE 1 + 2 = 3 and 3 + 4 = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q3/5

True or false?

Multiplying a 2-by-2 matrix by a 2-by-1 column gives a 2-by-1 column.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q4/5

966

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(2 \times 2)(2 \times 1)$ gives 2×1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q4/5

True or false?

For all 2-by-2 matrices A and B, AB = BA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q5/5

967

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Matrix multiplication is not commutative in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q5/5

Solve systems of linear equations in context, including systems with no solution or infinitely many solutions

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Write and solve an equation for one quantity in context	**HSA-CED.A.1**
BACKWARD 2	Solve real-world problems with two linear equations	**8.EE.C.8.c**
FORWARD 1	Use matrices to organize data in context	**HSN-VM.C.6**
FORWARD 2	Solve a system in context with an inverse matrix	**HSA-REI.C.9**

SET 28 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

A theater sold 340 tickets. Adult tickets cost $12 and student tickets cost $7. Sales totaled $3,255.

Write a system and find how many of each kind were sold.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q1/5

970

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

a + s = 340 and 12a + 7s = 3,255; 175 adult and 165 student tickets

GRADING NOTE Form required: a system and both numbers. 12a + 7(340 − a) = 3,255 gives 5a = 875. Check: 2,100 + 1,155 = 3,255.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q1/5

How many solutions does the system $6x - 4y = 10$ and $3x - 2y = 5$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q2/5

971

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

Infinitely many. The first equation is 2 times the second, so both describe the same line.

GRADING NOTE Form required: the number of solutions and a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q2/5

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 liters of a 25% acid solution.

How many liters of each are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q3/5

QUESTION ID

S28-M-Q3

TYPE

Short answer

CORRECT ANSWER

5 liters of the 10% solution and 15 liters of the 30% solution

GRADING NOTE Form required: both amounts, labeled. $x + y = 20$ and $0.10x + 0.30y = 5$. Then $6 - 0.2x = 5$, so $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q3/5

For which value of k does the system y = 3x + 2 and 6x − 2y = k have infinitely many solutions?

A. k = 2

B. k = −2

C. k = 4

D. k = −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q4/5

973

QUESTION ID

S28-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. k = −4

GRADING NOTE Substitute: 6x − 2(3x + 2) = −4 for every x, so the equations describe the same line only when k = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q4/5

True or false?

A system of two linear equations in two variables can have exactly two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q5/5

974

QUESTION ID
S28-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Two lines meet at no point, one point, or every point (the same line).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 1

Write and solve an equation for one quantity in context • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A gym has 340 members. There are 40 more adults than teens.

Write and solve an equation for the number of teens t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q1/5

976

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

t + (t + 40) = 340;
t = 150

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Pens cost $2 and notebooks cost $5. Ana buys 3 notebooks and some pens and spends $27.

Write and solve an equation for the number of pens p.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q2/5

977

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2p + 15 = 27; p = 6

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q2/5

A 20-liter mix has x liters of a 10% acid solution and the rest a 30% acid solution.

Which expression gives the liters of acid?

A. 0.40(20)

B. 0.10x + 0.30x

C. 0.10x + 0.30(20 − x)

D. 10x + 30(20 − x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q3/5

978

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C.

0.10x + 0.30(20 − x)

GRADING NOTE The rest of the mix is 20 − x liters; 10% = 0.10 and 30% = 0.30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q3/5

True or false?

25 coins are all dimes and quarters. If d coins are dimes, then 25 − d coins are quarters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q4/5

979

QUESTION ID
S28-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The two counts add to 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

"3 more than 5 times a number is 48" gives 5n + 3 = 48, and its solution is n = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q5/5

980

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5n = 45, so n = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 2

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

ANSWER KEY Reduced copy of the question page, with its answer

Two numbers have a sum of 30 and a difference of 8.

Find the numbers.

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q1/5

982

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

19 and 11

GRADING NOTE Any order. Adding the equations gives 2x = 38.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q1/5

Plan A costs $10 plus $2 per hour. Plan B costs $4 per hour.

After how many hours do they cost the same?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q2/5

983

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 10 + 2h = 4h, so h = 5 (both cost $20).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

3 pencils and 2 erasers cost $1.30. 1 pencil and 2 erasers cost $0.70.

What does one pencil cost?

A. $0.60

B. $0.20

C. $0.35

D. $0.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q3/5

984

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $0.30

GRADING NOTE Subtracting the equations gives 2 pencils = $0.60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q3/5

True or false?

Two cars leave the same place at the same time in the same direction, at 50 mi/h and 60 mi/h. After 2 hours the faster car is 20 miles ahead.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q4/5

985

QUESTION ID
S28-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $120 - 100 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q4/5

True or false?

One company charges y = 2x + 5 dollars and another charges y = 2x + 8 dollars for x units. They charge the same for 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q5/5

986

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The lines are parallel, so the costs are never equal; the second always costs $3 more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 1

Use matrices to organize data in context • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6

Friday: 120 adult and 90 student tickets. Saturday: 150 adult and 60 student tickets.

Write a 2-by-2 matrix with one row per day (adult first).

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q1/5

988

QUESTION ID
S28-F1-Q1

TYPE
Short answer

CORRECT ANSWER
[120 90] and [150 60]

GRADING NOTE Form required: Friday row first, adult column first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q1/5

Adult tickets cost $12 and student tickets $7.

Multiply the matrix with rows [120 90] and [150 60] by the column [12, 7]. What do the entries mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q2/5

989

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The column [2,070, 2,220]: Friday sales were $2,070 and Saturday sales were $2,220.

GRADING NOTE Form required: both entries and their meaning. 1,440 + 630 = 2,070 and 1,800 + 420 = 2,220.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q2/5

A matrix has rows [5 2] and [3 4]. Each row is a store and each column is a product.

What does the entry in row 2, column 1 represent?

A. Store 2's amount of product 1

B. Store 1's amount of product 2

C. Store 2's amount of product 2

D. The total of product 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q3/5

990

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Store 2's amount of product 1

GRADING NOTE Row 2 is store 2, and column 1 is product 1. The entry is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q3/5

True or false?

A matrix with 3 rows and 2 columns is a 3-by-2 matrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q4/5

991

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Dimensions are written rows by columns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q4/5

True or false?

Any two matrices can be added, whatever their dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q5/5

992

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only matrices with the same dimensions can be added.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 2

Solve a system in context with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

Write the system a + s = 340 and 12a + 7s = 3,255 as a matrix equation AX = B. Give A, X, and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q1/5

994

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A has rows [1 1] and [12 7]; X is the column [a, s]; B is the column [340, 3,255]; and AX = B.

GRADING NOTE Form required: A, X, and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q1/5

The matrix with rows [1 1] and [12 7] has determinant −5.

Does the ticket system have exactly one solution? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q2/5

995

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The determinant is not 0, so A has an inverse and $X = A^{-1}B$ is the only solution.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q2**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q2/5

The inverse of the matrix with rows [1 1] and [12 7] is (1 ÷ (−5)) times which matrix?

A. Rows [7 −1] and [−12 1]

B. Rows [7 1] and [12 1]

C. Rows [−7 1] and [12 −1]

D. Rows [1 −1] and [−12 7]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q3/5

996

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Rows [7 −1] and [−12 1]

GRADING NOTE For rows [a b] and [c d], the inverse is (1 ÷ (ad − bc)) times rows [d −b] and [−c a].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system has determinant 0, the system does not have exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q4/5

997

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The matrix has no inverse; the system has no solution or infinitely many.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q4**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q4/5

True or false?

The determinant of the matrix with rows [2 3] and [4 6] is 24.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q5/5

998

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2(6) - 3(4) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q5/5

Solve a system of a linear equation and a quadratic equation algebraically

MAIN	Algebra I main question	**HSA-REI.C.7**
BACKWARD 1	Solve quadratic equations by factoring	**HSA-REI.B.4.b**
BACKWARD 2	Solve systems of linear equations	**HSA-REI.C.6**
FORWARD 1	Decide whether points lie on a circle	**HSG-GPE.B.4**
FORWARD 2	Find complex solutions when a line and a parabola do not meet	**HSN-CN.C.7**

SET 29 • HSA-REI.C.7

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system.

$y = x^2 - 3$

$y = 2x$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q1/5

1001

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 6) and (−1, −2)

GRADING NOTE Form required: both ordered pairs. $x^2 - 3 = 2x$ gives $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system.

$y = x^2 + 4x + 1$

$y = x - 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q2/5

1002

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, −2) and (−2, −3)

GRADING NOTE Form required: both ordered pairs. $x^2 + 3x + 2 = (x + 1)(x + 2) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many real solutions does the system $y = x^2 + 2$ and $y = x$ have?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q3/5

1003

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

GRADING NOTE $x^2 - x + 2 = 0$ has discriminant $1 - 8 = -7 < 0$, so the line and the parabola do not meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system.

$x^2 + y^2 = 25$

$y = x + 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q4/5

1004

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

(3, 4) and (−4, −3)

GRADING NOTE Form required: both ordered pairs. $x^2 + (x + 1)^2 = 25$ gives $x^2 + x - 12 = 0$, so $x = 3$ or $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q4/5

True or false?

The line $y = 2x - 1$ meets the parabola $y = x^2$ at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q5/5

1005

QUESTION ID

S29-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 - 2x + 1 = (x - 1)^2 = 0$ has one solution, $x = 1$. The line touches the parabola at (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 2x - 3 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q1/5

1007

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 3 or x = −1

GRADING NOTE Form required: both solutions. (x − 3)(x + 1) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + x - 12 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q2/5

1008

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 or x = −4

GRADING NOTE Form required: both solutions. (x + 4)(x − 3) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 3x + 2 = 0$.

A. $x = -1$ or $x = 2$

B. $x = 1$ or $x = 2$

C. $x = 1$ or $x = -2$

D. $x = -1$ or $x = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q3/5

1009

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = -1$ or $x = -2$

GRADING NOTE $(x + 1)(x + 2) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 - x + 2 = 0$ has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q4/5

1010

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The discriminant is $1 - 8 = -7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q4/5

1010

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The solutions of $x^2 = 2x$ are $x = 2$ and $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q5/5

1011

QUESTION ID
S29-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x(x - 2) = 0$, so $x = 0$ or $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 2

Solve systems of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$y = 2x$

$y = -x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q1/5

1013

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 6)

GRADING NOTE Any equivalent form. $2x = -x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + y = 1$

$y = x + 1$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q2/5

1014

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent form. $x + x + 1 = 1$, so $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$2x + y = 7$

$x - y = 2$

A. (4, −1)

B. (3, 1)

C. (1, 3)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q3/5

1015

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 1)

GRADING NOTE Adding gives 3x = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q3/5

True or false?

Substitution and elimination give the same solution of a system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q4/5

1016

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both methods keep the solutions unchanged at every step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q4/5

True or false?

(2, 1) is a solution of the system $x + y = 3$ and $x - y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q5/5

1017

QUESTION ID
S29-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2 - 1 = 1$, not 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSA-REI.C.7

FORWARD 1

Decide whether points lie on a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A circle is centered at the origin and passes through (0, 5).

Does the point (3, −4) lie on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q1/5

1019

QUESTION ID
S29-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. The radius is 5, and the distance from (0, 0) to (3, −4) is √(9 + 16) = 5.

GRADING NOTE Any equivalent explanation using distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the points where the line $y = 3$ meets the circle $x^2 + y^2 = 25$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q2/5

1020

QUESTION ID
S29-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(4, 3) and (−4, 3)

GRADING NOTE Form required: both points. $x^2 + 9 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (1, 2) and radius 5.

Which point is on the circle?

A. (1, 6)

B. (4, 6)

C. (3, 3)

D. (6, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q3/5

1021

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (4, 6)

GRADING NOTE From (1, 2) to (4, 6): √(9 + 16) = 5. (3, 3) is √5 away, (6, 3) is √26 away, and (1, 6) is 4 away.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q3/5

True or false?

The line $y = 6$ does not meet the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q4/5

1022

QUESTION ID
S29-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The circle reaches only up to $y = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The point (2, 2) lies on the circle $x^2 + y^2 = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q5/5

1023

QUESTION ID
S29-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 4 + 4 = 8, not 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

FORWARD 2

Find complex solutions when a line and a parabola do not meet • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Substituting $y = 2x$ into $y = x^2 + 2$ gives $x^2 - 2x + 2 = 0$.

Solve it over the complex numbers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q1/5

1025

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 1 + i$ or $x = 1 - i$

GRADING NOTE Form required: a + bi (both solutions). $x = (2 \pm \sqrt{(-4)}) \div 2$. The solutions are not real, so the graphs do not meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 4 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q2/5

1026

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 2i$ or $x = -2i$

GRADING NOTE Form required: both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 6x + 10 = 0$.

A. $3 + i$ and $3 - i$

B. $6 + i$ and $6 - i$

C. $3 + 2i$ and $3 - 2i$

D. $-3 + i$ and $-3 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q3/5

1027

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3 + i and 3 − i

GRADING NOTE $(x - 3)^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q3/5

True or false?

When a line and a parabola do not intersect, the equation from substitution has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q4/5

1028

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real solution x would give a common point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q4/5

True or false?

$i^3 = i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q5/5

1029

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^3 = i^2 \cdot i = -i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q5/5

Understand the graph of an equation as the set of all its solutions

MAIN	Algebra I main question	**HSA-REI.D.10**
BACKWARD 1	Use tables and graphs to relate two variables	**6.EE.C.9**
BACKWARD 2	Understand a function's graph as its input-output pairs	**8.F.A.1**
FORWARD 1	Derive a parabola from its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use the unit circle as the set of points (cos θ, sin θ)	**HSF-TF.A.2**

SET 30 • HSA-REI.D.10

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (3, −2) on the graph of 2x + 5y = −4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q1/5

1032

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(3) + 5(−2) = 6 − 10 = −4, so (3, −2) is a solution.

GRADING NOTE Any equivalent explanation that substitutes the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

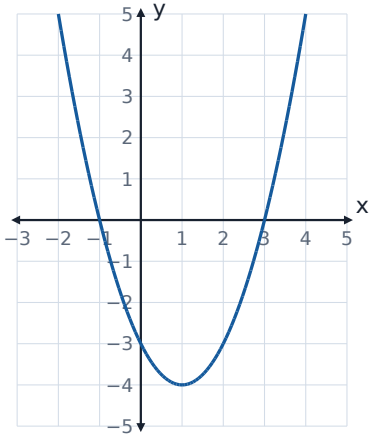
High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q1/5

The graph of $y = x^2 - 2x - 3$ is shown.

Which point is NOT a solution of the equation?

A. (2, 3) **B.** (3, 0)

C. (0, −3) **D.** (1, −4)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q2/5

1033

QUESTION ID

S30-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 3)

GRADING NOTE At $x = 2$, $y = 4 - 4 - 3 = -3$, not 3. The other points are on the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q2/5

The point (k, 7) is on the graph of y = 3x − 2.

Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q3/5

1034

QUESTION ID
S30-M-Q3

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. 3k − 2 = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q3/5

The graph of an equation in x and y passes through (4, 1) and (−2, 5).

What does that tell you about the equation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q4/5

1035

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 4, y = 1 and x = −2, y = 5 both make the equation true: each is a solution.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q4/5

True or false?

A point that is not on the graph of $y = x^2$ can still be a solution of $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q5/5

1036

QUESTION ID
S30-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The graph is exactly the set of all solutions, so every solution is on it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 1

Use tables and graphs to relate two variables • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

y = 3x. Find y for x = 0, 1, 2, and 3.

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q1/5

1038

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 3, 6, 9

GRADING NOTE Any equivalent form, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A car travels at 40 miles per hour.

Write an equation for the distance d after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q2/5

1039

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

d = 40t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the graph of y = x + 4?

A. (2, 6)

B. (1, 3)

C. (4, 0)

D. (6, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q3/5

1040

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 6)

GRADING NOTE 2 + 4 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = 5x, y is the dependent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q4/5

1041

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE y depends on the value chosen for x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (2, 8) is on the graph of y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q5/5

1042

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3(2) = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 2

Understand a function's graph as its input-output pairs • Grade 8 • Functions • 8.F.A.1

ANSWER KEY Reduced copy of the question page, with its answer

The function f pairs each x with 2x − 1.

What output does it pair with x = 4?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q1/5

1044

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is {(1, 2), (2, 3), (1, 5)} a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q2/5

1045

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No. The input 1 is paired with two outputs, 2 and 5.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q2/5

Which point is on the graph of $y = x^2$?

A. (9, −3)

B. (−3, −9)

C. (−3, 9)

D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q3/5

1046

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (−3, 9)

GRADING NOTE $(-3)^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q4/5

1047

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition of a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The vertical line x = 2 is the graph of a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q5/5

1048

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The input 2 would have infinitely many outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 1

Derive a parabola from its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A parabola is the set of points equally far from the focus (0, 2) and the directrix $y = -2$.

Derive its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q1/5

1050

QUESTION ID

S30-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x^2 = 8y$

GRADING NOTE REQUIRED WORK: The distance condition $\sqrt{(x^2 + (y - 2)^2)} = |y + 2|$ written, both sides squared, and simplified to $x^2 = 8y$.

Form required: the derivation and the equation (any equivalent form, such as $y = x^2 \div 8$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q1/5

Is (4, 2) on the parabola with focus (0, 2) and directrix $y = -2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q2/5

1051

QUESTION ID
S30-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes. It is 4 units from the focus and 4 units from the directrix.

GRADING NOTE Any equivalent explanation (or $4^2 = 8 \cdot 2$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q2/5

Where is the focus of the parabola $y = x^2 \div 4$?

A. (0, 4)

B. $(0, \frac{1}{4})$

C. (1, 0)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q3/5

1052

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 1)

GRADING NOTE $x^2 = 4y = 4py$ with $p = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q3**

**High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q3/5**

True or false?

Every point on a parabola is the same distance from its focus as from its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q4/5

1053

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition of a parabola.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The vertex of a parabola lies on its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q5/5

1054

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The vertex is halfway between the focus and the directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 2

Use the unit circle as the set of points (cos θ, sin θ) • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

Explain why the point $(\cos\theta, \sin\theta)$ lies on the circle $x^2 + y^2 = 1$ for every angle θ.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q1/5

1056

QUESTION ID

S30-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\cos^2\theta + \sin^2\theta = 1$, so the point satisfies the equation. (It is 1 unit from the origin.)

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q1/5

Which point on the unit circle corresponds to θ = π ÷ 2?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q2/5

1057

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent form. cos(π ÷ 2) = 0 and sin(π ÷ 2) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q2/5

Which point is on the unit circle $x^2 + y^2 = 1$?

A. (0.5, 0.5)

B. (1, 1)

C. (0.6, 0.8)

D. (0.8, 0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q3/5

1058

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (0.6, 0.8)

GRADING NOTE 0.36 + 0.64 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q3/5

True or false?

The point $(-1, 0)$ on the unit circle corresponds to an angle of π radians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q4/5

1059

QUESTION ID
S30-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $\cos \pi = -1$ and $\sin \pi = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The point (cos 0, sin 0) is (0, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q5/5

1060

QUESTION ID
S30-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE cos 0 = 1 and sin 0 = 0, so the point is (1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q5/5

Explain why intersections of y = f(x) and y = g(x) solve f(x) = g(x), and approximate solutions with graphs and tables

MAIN	Algebra I main question	**HSA-REI.D.11**
BACKWARD 1	Check whether points are on a graph	**HSA-REI.D.10**
BACKWARD 2	Read the solution of a system from the graphs of two lines	**8.EE.C.8.a**
FORWARD 1	Solve exponential equations exactly with logarithms	**HSF-LE.A.4**
FORWARD 2	Solve trigonometric equations with graphs and technology	**HSF-TF.B.7**

SET 31 • HSA-REI.D.11

MAIN

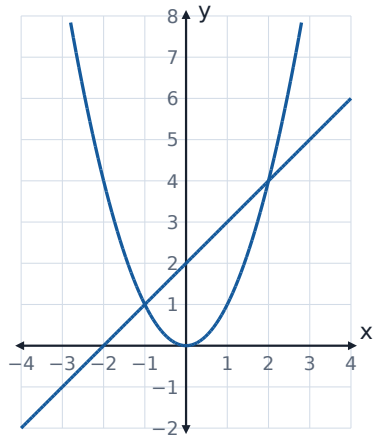
Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ are shown.

Use the graphs to solve $x^2 = x + 2$.

Solutions: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q1/5

1063

QUESTION ID

S31-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x = -1$ or $x = 2$

GRADING NOTE Form required: both solutions. The graphs meet at $(-1, 1)$ and $(2, 4)$. Check: $(-1)^2 = 1 = -1 + 2$ and $2^2 = 4 = 2 + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a table of values to find the solution of $2^x = 3x$ between $x = 3$ and $x = 4$, to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q2/5

1064

QUESTION ID

S31-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

3.3

GRADING NOTE

REQUIRED WORK: A table of 2^x and 3x at x-values that bracket the solution closely enough to decide the tenth (for example x = 3.3 and x = 3.35), showing where $2^x - 3x$ changes sign.

Form required: rounded to the nearest tenth, with the table. At 3.3, $2^x \approx 9.85 < 9.9 = 3x$; at 3.35, $2^x \approx 10.20 > 10.05 = 3x$, so the solution (about 3.31) rounds to 3.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows f(x) = |x − 1| and g(x) = 3 − x.

For which x is f(x) = g(x)?

x	0	1	2	3	4
f(x)	1	0	1	2	3
g(x)	3	2	1	0	−1

A. x = 2 **B.** x = 3 **C.** x = 0 **D.** x = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q3/5

1065

QUESTION ID

S31-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 2

GRADING NOTE At x = 2 both equal 1. In every other column the outputs differ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the x-coordinates of the points where the graphs of y = f(x) and y = g(x) meet are the solutions of f(x) = g(x).

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q4/5

1066

QUESTION ID

S31-M-Q4

TYPE

Short answer

CORRECT ANSWER

At a meeting point (a, b), b = f(a) and b = g(a), so f(a) = g(a). If f(a) = g(a), the point (a, f(a)) is on both graphs.

GRADING NOTE Any equivalent explanation that uses the coordinates of a point on both graphs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q4/5

True or false?

The graphs of $y = 3x + 1$ and $y = 3x - 4$ never meet, so $3x + 1 = 3x - 4$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q5/5

1067

QUESTION ID
S31-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The lines are parallel, and $1 = -4$ is false for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.D.11

BACKWARD 1

Check whether points are on a graph • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 4) on both $y = x^2$ and $y = x + 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q1/5

1069

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. $2^2 = 4$ and $2 + 2 = 4$.

GRADING NOTE Any equivalent explanation that checks both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point $(-1, y)$ is on the graph of $y = x + 2$.

Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q2/5

1070

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the graph of $y = |x - 1|$?

A. $(-2, -3)$

B. $(-2, 3)$

C. $(3, -2)$

D. $(1, 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q3/5

1071

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−2, 3)

GRADING NOTE $|-2 - 1| = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If (a, b) is on the graph of y = f(x), then b = f(a).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q4/5

1072

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph is the set of points (x, f(x)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(0, 1) is on the graph of $y = 2^x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q5/5

1073

QUESTION ID
S31-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2^0 + 1 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSA-REI.D.11

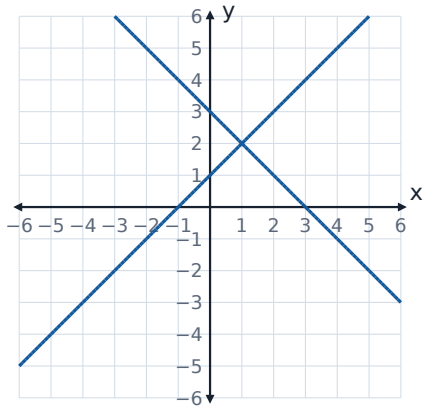
BACKWARD 2

Read the solution of a system from the graphs of two lines • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Use the graph to solve the system $y = x + 1$ and $y = -x + 3$.

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q1/5

1075

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(1, 2)

GRADING NOTE Any equivalent form. The lines cross at (1, 2). Check: 1 + 1 = 2 and −1 + 3 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

At what x-value do the graphs of $y = 2x - 3$ and $y = 5$ meet?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q2/5

1076

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $2x - 3 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q2/5

The graphs of two linear equations are the same line.

How many solutions does the system have?

A. One

B. Infinitely many

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q3/5

1077

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Infinitely many

GRADING NOTE Every point of the line is on both graphs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q3/5

True or false?

Two lines with different slopes meet at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q4/5

1078

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lines with different slopes are not parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q4/5

True or false?

Two lines with the same slope always meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q5/5

1079

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Distinct lines with the same slope are parallel and never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 1

Solve exponential equations exactly with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 2^x$ and $y = 10$ meet at one point.

Find its x-coordinate exactly using logarithms, and to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q1/5

1081

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = log 10 ÷ log 2 = 1 ÷ log 2 ≈ 3.32

GRADING NOTE Form required: the exact expression (such as $\log_2 10$) and the rounded value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5e^x = 20$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q2/5

1082

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

ln 4

GRADING NOTE Form required: exact. $e^x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph shows $y = 3^x$ meeting $y = 15$ at about $x = 2.5$.

Which expression is the exact solution of $3^x = 15$? (log means base 10.)

A. $15 \div 3$

B. log 5

C. log 15 ÷ log 3

D. $\sqrt{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q3/5

1083

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. log 15 ÷ log 3

GRADING NOTE Take the logarithm of both sides: x log 3 = log 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log_2 8 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q4/5

1084

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2^3 = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The solution of $10^x = 0.01$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q5/5

1085

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $0.01 = 10^{-2}$, so $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 2

Solve trigonometric equations with graphs and technology • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sin x = \frac{1}{2}$ for $0 \le x \le \pi$. Give exact answers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q1/5

1087

QUESTION ID

S31-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x = π ÷ 6 or

x = 5π ÷ 6

GRADING NOTE Form required: exact, both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q1/5

Use a calculator to solve $\cos x = 0.3$ for $0 \le x \le \pi$, to the nearest hundredth of a radian.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q2/5

1088

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.27

GRADING NOTE Form required: rounded to the nearest hundredth. $x = \cos^{-1}(0.3) \approx 1.266$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many solutions does sin x = 2 have?

A. Two

B. One

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q3/5

1089

QUESTION ID
S31-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. None

GRADING NOTE The values of sin x are between −1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q3/5

True or false?

The graphs of y = sin x and y = 0.5 meet at infinitely many points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q4/5

1090

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The sine graph repeats every 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q4/5

True or false?

For $0 \le x \le \pi$, the equation $\cos x = 0$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q5/5

1091

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only $x = \pi \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q5/5

Solve linear-quadratic systems by graphing and check the solutions algebraically

MAIN	Algebra I main question	**HSA-REI.C.7**
BACKWARD 1	Read solutions of f(x) = g(x) from intersections	**HSA-REI.D.11**
BACKWARD 2	Check whether a point is on a graph	**HSA-REI.D.10**
FORWARD 1	Find where a line meets a circle with coordinates	**HSG-GPE.B.4**
FORWARD 2	Find where a line meets an ellipse	**HSG-GPE.A.3**

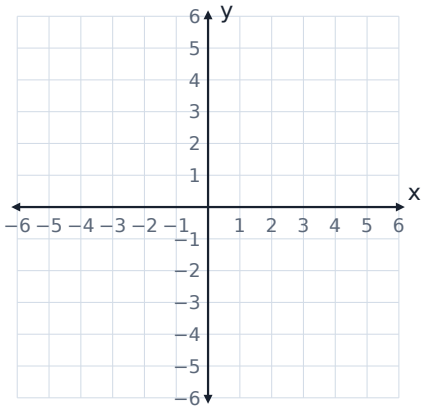
SET 73 • HSA-REI.C.7

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Graph $y = x^2 - 2$ and $y = x$ on the grid. Read the solutions of the system from the intersections, then check them algebraically.



Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q1/5

1094

QUESTION ID

S73-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The parabola $y = x^2 - 2$ (vertex $(0, -2)$) and the line $y = x$, crossing at $(-1, -1)$ and $(2, 2)$

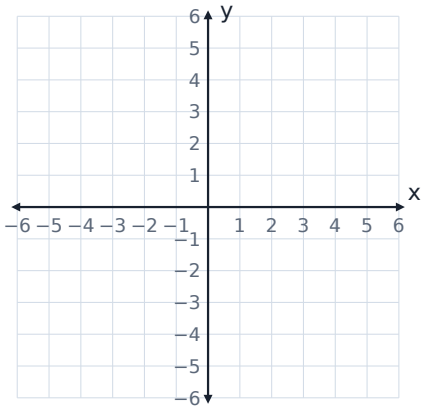
Written: $(-1, -1)$ and $(2, 2)$

GRADING NOTE Form required: both points with an algebraic check: $x^2 - 2 = x$ gives $(x - 2)(x + 1) = 0$. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q1/5

Graph the circle $x^2 + y^2 = 25$ and the line $y = x + 1$. Read the intersection points, then check one of them algebraically.



Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q2/5

1095

QUESTION ID

S73-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A circle of radius 5 centered at the origin and the line through (0, 1) with slope 1, crossing at (3, 4) and (−4, −3)

Written: (3, 4) and (−4, −3)

GRADING NOTE Form required: both points and one algebraic check, such as 9 + 16 = 25 and 4 = 3 + 1.

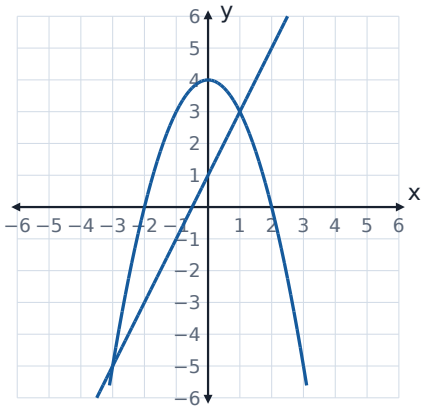
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = -x^2 + 4$ and $y = 2x + 1$ are shown.

Read the solutions of the system from the graph, then verify them algebraically.



Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q3/5

1096

QUESTION ID

S73-M-Q3

TYPE

Short answer

CORRECT ANSWER

(1, 3) and (−3, −5)

GRADING NOTE Form required: both points with the check $-x^2 + 4 = 2x + 1$, so $x^2 + 2x - 3 = (x + 3)(x - 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q3/5

The graphs of $y = x^2$ and $y = 2x - 1$ are shown.

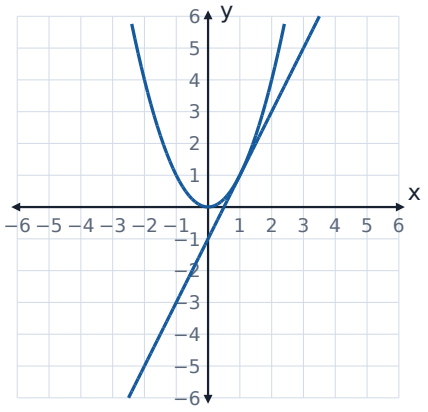
How many solutions does the system have?

A. One

B. Two

C. Infinitely many

D. None



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q4/5

1097

QUESTION ID

S73-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. One

GRADING NOTE $x^2 = 2x - 1$ gives $(x - 1)^2 = 0$, so the line touches the parabola only at (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q4/5

True or false?

The line $y = 5$ and the circle $x^2 + y^2 = 9$ intersect at two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q5/5

1098

QUESTION ID

S73-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The circle has radius 3, so it reaches only up to $y = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q5/5

SET 73 • HSA-REI.C.7

BACKWARD 1

Read solutions of f(x) = g(x) from intersections • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = x^2$ and $y = 4$ meet at two points.
What are the solutions of $x^2 = 4$?

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q1/5

1100

QUESTION ID
S73-B1-Q1

TYPE
Short answer

CORRECT ANSWER
x = 2 or x = −2

GRADING NOTE Form required: both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 2x + 1$ and $y = x^2 + 1$ meet at (0, 1) and (2, 5). Which x-values solve $2x + 1 = x^2 + 1$? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q2/5

1101

QUESTION ID

S73-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 0 and x = 2: at those x-values the two expressions have the same output.

GRADING NOTE Form required: both values and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of y = f(x) and y = g(x) meet only at (3, 7). Which statement is true?

A. f(x) = g(x) for every x

B. f(7) = g(7) = 3

C. f(3) = 3

D. f(3) = g(3) = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q3/5

1102

QUESTION ID

S73-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. f(3) = g(3) = 7

GRADING NOTE An intersection point (a, b) means f(a) = g(a) = b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q3/5

True or false?

If the graphs of y = f(x) and y = g(x) never meet, then f(x) = g(x) has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q4/5

1103

QUESTION ID

S73-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A solution would give a common point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q4/5

True or false?

The solutions of f(x) = g(x) are the y-coordinates of the intersection points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q5/5

1104

QUESTION ID

S73-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are the x-coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSA-REI.C.7

BACKWARD 2

Check whether a point is on a graph • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (3, 4) on the circle $x^2 + y^2 = 25$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q1/5

1106

QUESTION ID

S73-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 9 + 16 = 25.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 3) on the parabola $y = x^2 - 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q2/5

1107

QUESTION ID

S73-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No. $2^2 - 2 = 2$, not 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on both $y = x$ and $y = x^2 - 2$?

A. (2, 2)

B. (−2, −2)

C. (1, 1)

D. (0, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q3/5

1108

QUESTION ID

S73-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 2)

GRADING NOTE $2 = 2$ and $2^2 - 2 = 2$. Each other point fails one of the equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q3/5

True or false?

A point is a solution of a system exactly when it lies on the graphs of both equations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q4/5

1109

QUESTION ID
S73-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each graph is the set of solutions of its equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(0, 5) is on both the circle $x^2 + y^2 = 25$ and the line $y = x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q5/5

1110

QUESTION ID
S73-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is on the circle, but 0 + 1 = 1, not 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSA-REI.C.7

FORWARD 1

Find where a line meets a circle with coordinates • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the points where the line $y = 7 - x$ meets the circle $x^2 + y^2 = 25$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q1/5

1112

QUESTION ID
S73-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(3, 4) and (4, 3)

GRADING NOTE Form required: both points. $x^2 + (7 - x)^2 = 25$ gives $x^2 - 7x + 12 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q1/5

Show that the line $y = 5$ touches the circle $x^2 + y^2 = 25$ at exactly one point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S73-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q2/5

1113

QUESTION ID

S73-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 25 = 25$ gives $x^2 = 0$, so the only point is (0, 5).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q2/5

How many points do the line $y = 2x$ and the circle $x^2 + y^2 = 20$ share?

A. One

B. Four

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q3/5

1114

QUESTION ID
S73-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. Two

GRADING NOTE $x^2 + 4x^2 = 20$ gives $x = 2$ or $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q3/5

True or false?

The line $x = 6$ does not meet the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q4/5

1115

QUESTION ID
S73-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The circle reaches only to $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q4/5

True or false?

The point (1, 5) lies on the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q5/5

1116

QUESTION ID
S73-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $1 + 25 = 26$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSA-REI.C.7

FORWARD 2

Find where a line meets an ellipse • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find where the line $y = 2$ meets the ellipse $x^2 \div 16 + y^2 \div 4 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q1/5

1118

QUESTION ID
S73-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Only at (0, 2): the line touches the ellipse there.

GRADING NOTE Form required: the point. $x^2 \div 16 + 1 = 1$ gives $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find where the x-axis meets the ellipse $x^2 \div 9 + y^2 \div 16 = 1$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q2/5

1119

QUESTION ID
S73-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(3, 0) and (−3, 0)

GRADING NOTE Form required: both points. $y = 0$ gives $x^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q2/5

Where does the line x = 0 meet the ellipse $x^2 \div 25 + y^2 \div 9 = 1$?

A. (0, 5) and (0, −5)

B. (5, 0) and (−5, 0)

C. Nowhere

D. (0, 3) and (0, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q3/5

1120

QUESTION ID
S73-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. (0, 3) and (0, −3)

GRADING NOTE x = 0 gives $y^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q3/5

True or false?

A line can meet an ellipse in at most two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q4/5

1121

QUESTION ID
S73-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Substituting a line into the ellipse gives an equation of degree at most 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q4/5

True or false?

The line $y = 4$ meets the ellipse $x^2 \div 25 + y^2 \div 9 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q5/5

1122

QUESTION ID
S73-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The ellipse reaches only up to $y = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q5/5

Solve systems of linear equations by graphing, including approximate solutions

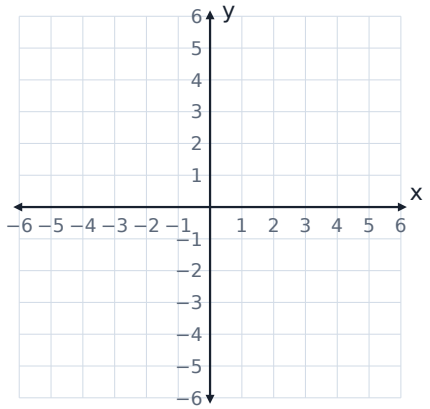
MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Find the solution of a system where two lines cross	**8.EE.C.8.a**
BACKWARD 2	Decide whether points are on a line	**HSA-REI.D.10**
FORWARD 1	Graph parallel and perpendicular lines and find where they meet	**HSG-GPE.B.5**
FORWARD 2	Check a graphical estimate with an inverse matrix	**HSA-REI.C.9**

SET 84 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Graph $y = 2x - 1$ and $y = -x + 5$ on the grid. What is the solution of the system?



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q1/5

1125

QUESTION ID

S84-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The line through (0, −1) with slope 2 and the line through (0, 5) with slope −1, crossing at (2, 3)

Written: (2, 3)

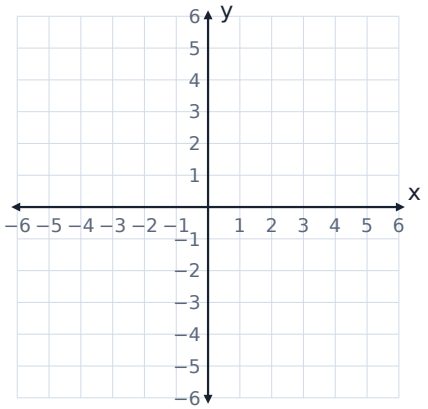
GRADING NOTE Form required: the solution read from the graph (it may be written x = 2, y = 3). Check: 4 − 1 = 3 and −2 + 5 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph y = 0.5x + 2 and y = −2x + 4. Estimate the solution to the nearest half unit, then solve algebraically to check.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q2/5

1126

QUESTION ID

S84-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The two lines crossing just left of x = 1 and just below y = 2.5

Written: Estimate about (1, 2.5); exact solution (0.8, 2.4)

GRADING NOTE Form required: an estimate within 0.5 of (0.8, 2.4) and the exact solution. 0.5x + 2 = −2x + 4 gives 2.5x = 2.

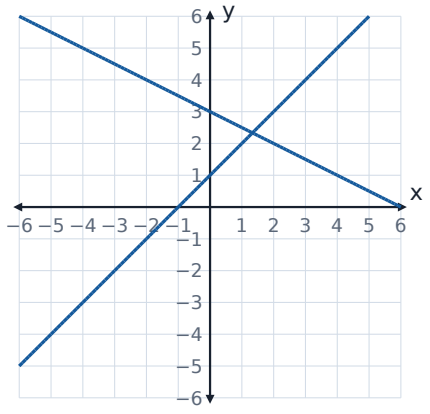
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = x + 1$ and $y = -0.5x + 3$ are shown.

Estimate the solution of the system to the nearest tenth, and explain how you made the estimate.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q3/5

1127

QUESTION ID

S84-M-Q3

TYPE

Short answer

CORRECT ANSWER

About (1.3, 2.3): the crossing is about one third of the way from x = 1 to x = 2 and from y = 2 to y = 3.

GRADING NOTE Form required: an estimate within 0.2 of the exact (4/3, 7/3) and a reason from the graph.

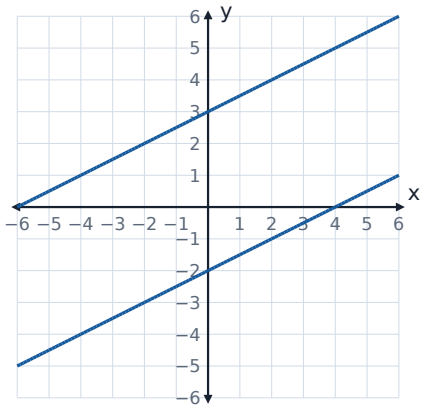
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of the two equations of a system are shown.

How many solutions does the system have?



A. Two

B. One

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q4/5

1128

QUESTION ID

S84-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. None

GRADING NOTE The lines are parallel (same slope, different intercepts), so they never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q4/5

True or false?

If the graphs of two linear equations lie on top of each other, the system has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q5/5

1129

QUESTION ID

S84-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every point of the line is on both graphs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q5/5

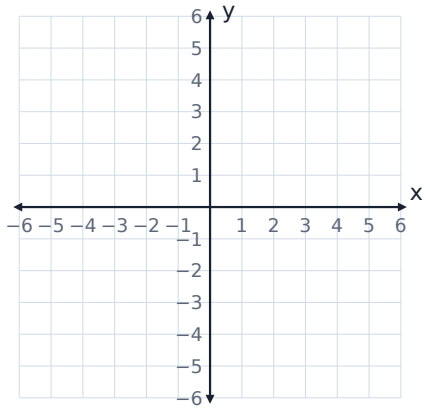
SET 84 • HSA-REI.C.6

BACKWARD 1

Find the solution of a system where two lines cross • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph y = x and y = 4 − x, and mark the point where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q1/5

1131

QUESTION ID

S84-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Two lines crossing at (2, 2), marked

GRADING NOTE Any correct graph with (2, 2) marked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines meet at (−1, 3). What is the solution of their system?

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q2/5

1132

QUESTION ID

S84-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, 3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q2/5

The graphs of two lines cross at (4, −2). Which equations could they be?

A. y = x − 6 and y = x + 2

B. y = x − 6 and y = −x + 2

C. y = x + 6 and y = −x + 2

D. y = 2x and y = −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q3/5

1133

QUESTION ID

S84-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = x − 6 and y = −x + 2

GRADING NOTE 4 − 6 = −2 and −4 + 2 = −2. In each other pair, (4, −2) fails at least one equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q3/5

True or false?

A point that lies on both lines is a solution of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q4/5

1134

QUESTION ID

S84-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It makes both equations true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q4/5

True or false?

Two lines with different slopes can have no point in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q5/5

1135

QUESTION ID

S84-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Lines with different slopes always cross exactly once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSA-REI.C.6

BACKWARD 2

Decide whether points are on a line • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 3) on the line y = 2x − 1? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q1/5

1137

QUESTION ID

S84-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(2) − 1 = 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 3) on the line y = −x + 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q2/5

1138

QUESTION ID

S84-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No. −2 + 4 = 2, not 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the line y = 0.5x + 2?

A. (2, 3)

B. (3, 2)

C. (0, 0.5)

D. (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q3/5

1139

QUESTION ID

S84-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 3)

GRADING NOTE 0.5(2) + 2 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q3/5

True or false?

The graph of a linear equation in two variables is the set of all its solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q4/5

1140

QUESTION ID

S84-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each point on the line is a solution, and each solution is on the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(0, 4) is on the graph of $y = -2x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q5/5

1141

QUESTION ID

S84-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At x = 0, y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

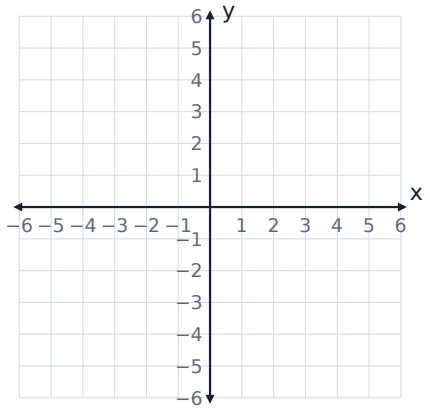
High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSA-REI.C.6

FORWARD 1

Graph parallel and perpendicular lines and find where they meet • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Graph y = 2x + 1 and the line through (4, 1) that is perpendicular to it. Mark the point where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S84-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q1/5

1143

QUESTION ID

S84-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The line y = −0.5x + 3 through (4, 1), crossing y = 2x + 1 at (0.8, 2.6)

GRADING NOTE Any correct graph. The perpendicular slope is $-\frac{1}{2}$; 2x + 1 = −0.5x + 3 gives x = 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S84-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q1/5

Two lines have slopes 3 and $-\frac{1}{3}$. What angle do they form?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q2/5

1144

QUESTION ID
S84-F1-Q2

TYPE
Short answer

CORRECT ANSWER
90°

GRADING NOTE Any equivalent form. The slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q2/5

Which system has no solution?

A. $y = 3x + 1$ and $y = -3x + 1$

B. $y = 3x + 1$ and $y = x + 1$

C. $y = 3x + 1$ and $y = 3x - 4$

D. $y = 3x$ and $y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q3/5

1145

QUESTION ID
S84-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $y = 3x + 1$ and $y = 3x - 4$

GRADING NOTE Same slope, different intercepts: parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q3/5

True or false?

The lines $y = -x + 2$ and $y = x - 2$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q4/5

1146

QUESTION ID
S84-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $(-1)(1) = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q4/5

True or false?

The lines y = 2x + 5 and y = 2x + 5 have no points in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q5/5

1147

QUESTION ID
S84-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They are the same line, so they share every point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSA-REI.C.6

FORWARD 2

Check a graphical estimate with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

The system y = 0.5x + 2 and y = −2x + 4 is written −0.5x + y = 2 and 2x + y = 4. Find the determinant of the matrix A with rows [−0.5 1] and [2 1].

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q1/5

1149

QUESTION ID
S84-F2-Q1

TYPE
Short answer

CORRECT ANSWER
−2.5

GRADING NOTE Any equivalent form. (−0.5)(1) − (1)(2) = −2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q1/5

A^{-1} is $(1 \div (-2.5))$ times the matrix with rows $[1\ -1]$ and $[-2\ -0.5]$. Use $X = A^{-1}B$, with B the column $[2, 4]$, to find the exact solution.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q2/5

1150

QUESTION ID

S84-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 0.8, y = 2.4

GRADING NOTE Form required: both values. $(1 \div (-2.5))$ times the column $[2 - 4, -4 - 2]$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q2**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q2/5

A graph shows two lines crossing near (1.3, 2.3). The inverse-matrix solution is (4/3, 7/3). Which statement is true?

A. The two results contradict each other.

B. The system has two solutions.

C. The graph is exact; the matrix method is approximate.

D. The graph gives an estimate; the matrix method gives the exact solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q3/5

1151

QUESTION ID

S84-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The graph gives an estimate; the matrix method gives the exact solution.

GRADING NOTE 4/3 ≈ 1.33 and 7/3 ≈ 2.33, which match the estimate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a 2-by-2 linear system has a nonzero determinant, the system has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q4/5

1152

QUESTION ID

S84-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then A has an inverse and $X = A^{-1}B$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q4/5

True or false?

If the graphs of two lines coincide, the determinant of the coefficient matrix is not 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q5/5

1153

QUESTION ID

S84-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Coinciding lines give a matrix with determinant 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q5/5

Graph the solutions of linear inequalities and systems of linear inequalities in two variables as half-planes

MAIN	Algebra I main question	**HSA-REI.D.12**
BACKWARD 1	Solve an inequality in two variables for y	**HSA-REI.B.3**
BACKWARD 2	Plot points and find distances in the coordinate plane	**6.NS.C.8**
FORWARD 1	Find the area of a region described by inequalities	**HSG-GPE.B.7**
FORWARD 2	Use zeros to find where a polynomial is positive or negative	**HSA-APR.B.3**

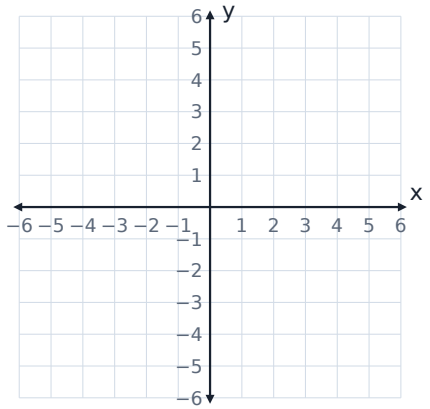
SET 32 • HSA-REI.D.12

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Graph $y < 2x - 1$ on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q1/5

1156

QUESTION ID

S32-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A dashed line through (0, −1) and (2, 3), with the half-plane below and to the right of it shaded

GRADING NOTE Any correct graph. The line is dashed because points on it are not solutions; (0, −2) is a solution and (0, 0) is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q1/5

Is (4, 1) a solution of the system $y \leq -x + 6$ and $y > \frac{1}{2}x - 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q2/5

1157

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. $1 \leq -4 + 6 = 2$, and $1 > 2 - 2 = 0$.

GRADING NOTE Any equivalent explanation that checks both inequalities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q2/5

A graph has a solid boundary line through (0, 2) and (3, 0), and the side that contains the origin is shaded.

Which inequality does it show?

A. $2x + 3y < 6$

B. $2x + 3y \geq 6$

C. $3x + 2y \leq 6$

D. $2x + 3y \leq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q3/5

1158

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2x + 3y \leq 6$

GRADING NOTE The line is $2x + 3y = 6$; solid means "or equal to"; (0, 0) gives $0 \leq 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

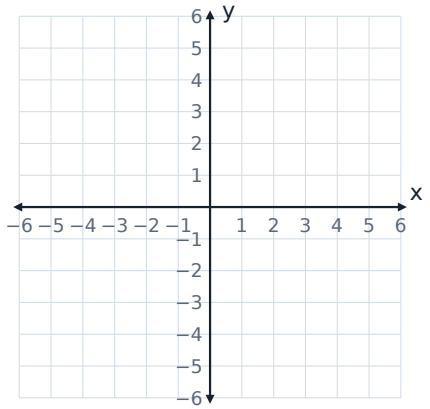
High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph the system $y \geq x - 2$ and $x + y < 4$ on the coordinate plane.

Name one point that is a solution of the system.

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q4/5

1159

QUESTION ID

S32-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A solid line $y = x - 2$ with the side above it shaded, a dashed line $x + y = 4$ with the side below it shaded, and the overlap marked

Written: Any point in the overlap, such as (0, 0)

GRADING NOTE Any equivalent form: the point must satisfy both inequalities. (0, 0): $0 \geq -2$ and $0 < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q4/5

True or false?

Points on the boundary line y = 3x are solutions of y > 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q5/5

1160

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE On the line, y = 3x, which is not greater than 3x. The boundary is dashed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 1

Solve an inequality in two variables for y • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 3y ≤ 6 for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q1/5

1162

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$y \leq -\frac{2}{3}x + 2$

GRADING NOTE Any equivalent form with y alone. 3y ≤ −2x + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x - 2y > 4$ for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q2/5

1163

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$y < \frac{1}{2}x - 2$

GRADING NOTE Any equivalent form with y alone. Dividing by −2 reverses the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-y \geq x - 5$ for y.

A. $y \geq x + 5$

B. $y \geq -x + 5$

C. $y \leq -x + 5$

D. $y \leq x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q3/5

1164

QUESTION ID
S32-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $y \leq -x + 5$

GRADING NOTE Multiply by −1 and reverse the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Solving $4x + y < 3$ for y gives $y < -4x + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q4/5

1165

QUESTION ID
S32-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Subtract 4x from both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q4/5

True or false?

Solving $-3y < 9$ for y gives $y < -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q5/5

1166

QUESTION ID
S32-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Dividing by −3 reverses the sign: $y > -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 2

Plot points and find distances in the coordinate plane • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (2, −3) and (2, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

1168

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 units

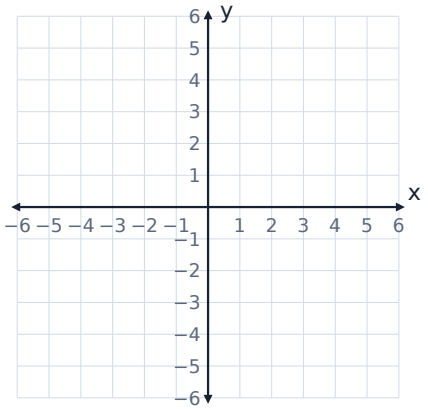
GRADING NOTE Any equivalent form. |5 − (−3)| = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Plot the points (−3, 2) and (4, −1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

1169

QUESTION ID

S32-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A point at (−3, 2) and a point at (4, −1)

GRADING NOTE Any correct plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which point is in Quadrant III?

A. (2, −5)

B. (−2, −5)

C. (2, 5)

D. (−2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

1170

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−2, −5)

GRADING NOTE Quadrant III has both coordinates negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

The points (3, 4) and (3, −4) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

1171

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Only the sign of y changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The distance between (−4, 1) and (6, 1) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

1172

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $|6 - (-4)| = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 1

Find the area of a region described by inequalities • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

The system $x \geq 0$, $y \geq 0$, and $x + y \leq 6$ describes a triangle.

Find its vertices and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q1/5

1174

QUESTION ID
S32-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(0, 0), (6, 0), and (0, 6); area 18 square units

GRADING NOTE Form required: the vertices and the area. $\frac{1}{2}(6)(6) = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q1/5

Find the perimeter of the rectangle with vertices (1, 1), (5, 1), (5, 4), and (1, 4).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q2/5

1175

QUESTION ID
S32-F1-Q2

TYPE
Short answer

CORRECT ANSWER
14 units

GRADING NOTE Any equivalent form. 2(4 + 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q2/5

The region $0 \le x \le 4$ and $0 \le y \le 2x$ is a triangle.

What is its area?

A. 32

B. 16

C. 8

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q3/5

1176

QUESTION ID
S32-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 16

GRADING NOTE Vertices (0, 0), (4, 0), and (4, 8): $\frac{1}{2}(4)(8) = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (3, 0), and (0, 4) has perimeter 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q4/5

1177

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3 + 4 + 5 = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q4/5

True or false?

The square with vertices (0, 0), (2, 0), (2, 2), and (0, 2) has area 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q5/5

1178

QUESTION ID
S32-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 2

Use zeros to find where a polynomial is positive or negative • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = (x - 1)(x - 4)$.

For which x is $p(x) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q1/5

1180

QUESTION ID
S32-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$1 < x < 4$

GRADING NOTE Form required: an inequality. Between the zeros, one factor is positive and one is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q1/5

$p(x) = x(x + 2)(x - 3)$.

Is $p(1)$ positive or negative? Explain using its factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q2/5

1181

QUESTION ID
S32-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Negative: $1 \cdot 3 \cdot (-2) = -6$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is $(x + 3)(x - 2) > 0$?

A. $x < -3$ only

B. $x > 2$ only

C. $x < -3$ or $x > 2$

D. $-3 < x < 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q3/5

1182

QUESTION ID
S32-F2-Q3

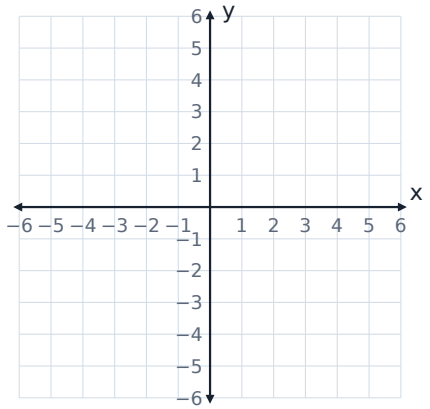
TYPE
Multiple choice

CORRECT ANSWER
C. $x < -3$ or $x > 2$

GRADING NOTE Both factors are positive when $x > 2$, and both are negative when $x < -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q3/5

Sketch a rough graph of $y = x(x + 2)(x - 3)$, and mark the x-values where $y < 0$. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S32-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q4/5

1183

QUESTION ID

S32-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −2, 0, and 3, falling to the left and rising to the right; $y < 0$ for $x < -2$ and for $0 < x < 3$

GRADING NOTE Any correct sketch with both negative intervals marked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S32-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 5)(x + 1) \geq 0$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q5/5

1184

QUESTION ID
S32-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE At x = 0 it equals −5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q5/5

Create equations in two or more variables and graph them on labeled axes

MAIN	Algebra I main question	**HSA-CED.A.2**
BACKWARD 1	Write and solve an equation in one variable	**HSA-CED.A.1**
BACKWARD 2	Write the equation of a linear function	**8.F.B.4**
FORWARD 1	Write the equation of a circle from a situation	**HSG-GPE.A.1**
FORWARD 2	Write and interpret polynomial models	**HSF-IF.C.7.c**

SET 33 • HSA-CED.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A savings account has $150 and grows by $25 each week.

Write an equation for the balance B after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q1/5

1187

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

B = 150 + 25w

GRADING NOTE Any equivalent form, such as B = 25w + 150.

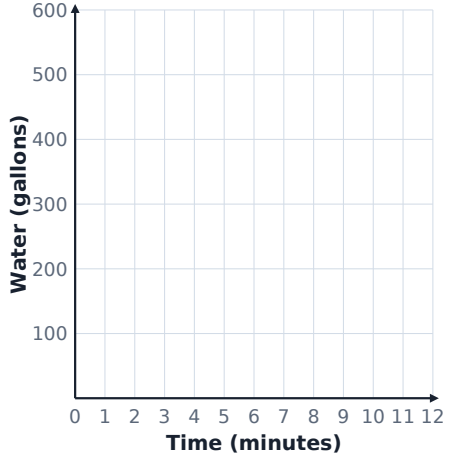
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q1/5

A pool holds 600 gallons and drains at 50 gallons per minute.

Write an equation for the gallons G left after t minutes, and graph it for 0 ≤ t ≤ 12.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q2/5

1188

QUESTION ID

S33-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line from (0, 600) to (12, 0)

Written:
G = 600 − 50t

GRADING NOTE Any equivalent form. Both the equation and the graph are graded. The pool is empty at t = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q2/5

A rectangle's length is 3 more than twice its width w.

Write an equation for its area A.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q3/5

1189

QUESTION ID

S33-M-Q3

TYPE

Short answer

CORRECT ANSWER

A = w(2w + 3)

GRADING NOTE Any equivalent form, such as $A = 2w^2 + 3w$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A town has 8,000 people, and its population grows 3% each year.

Which equation gives the population P after t years?

A. $P = 8{,}000(0.03)^t$

B. $P = 8{,}000 + 0.03t$

C. $P = 8{,}000(1.03)^t$

D. $P = 8{,}000(1.3)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q4/5

1190

QUESTION ID

S33-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $P = 8{,}000(1.03)^t$

GRADING NOTE Each year the population is multiplied by 1.03.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q4/5

True or false?

Student tickets cost $4 and adult tickets cost $9. The equation 4s + 9a = 300 relates the numbers of tickets when sales total $300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q5/5

1191

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 4s dollars from s student tickets plus 9a dollars from a adult tickets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q5/5

SET 33 • HSA-CED.A.2

BACKWARD 1

Write and solve an equation in one variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve 150 + 25w = 400 for w.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q1/5

1193

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. 25w = 250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q1/5

A pool has 600 gallons and drains at 50 gallons per minute.

Write and solve an equation for when it is empty.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q2/5

1194

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

600 − 50t = 0; t = 12 minutes

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A plan costs $12 per month plus a $30 fee.

Which equation gives the number of months m for a total of $186?

A. 12m − 30 = 186

B. 30m + 12 = 186

C. 12(m + 30) = 186

D. 12m + 30 = 186

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q3/5

1195

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12m + 30 = 186

GRADING NOTE 12 per month for m months, plus 30 once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q3/5

True or false?

w = 2.5 is a solution of w(2w + 3) = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q4/5

1196

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2.5(5 + 3) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q4/5

True or false?

Solving 4s + 9(20) = 300 gives s = 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q5/5

1197

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4s = 120, so s = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSA-CED.A.2

BACKWARD 2

Write the equation of a linear function • Grade 8 • Functions • 8.F.B.4

A line passes through (0, 5) and (2, 11).

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q1/5

1199

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 3x + 5

GRADING NOTE Any equivalent form. The slope is 6 ÷ 2 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plumber charges $50 plus $30 per hour.

Write a function for the cost C(h).

C(h) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q2/5

1200

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30h + 50

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q2/5

A table has x-values 0, 1, 2 and y-values 4, 7, 10.

Which equation fits the table?

A. y = 4x + 3

B. y = x + 4

C. y = 3x

D. y = 3x + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q3/5

1201

QUESTION ID

S33-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 3x + 4

GRADING NOTE y starts at 4 and increases by 3 for each 1 in x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q3/5

True or false?

The rate of change of $y = -2x + 7$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q4/5

1202

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The rate of change is the slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q4/5

True or false?

The initial value of $y = 6x - 1$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q5/5

1203

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value (at $x = 0$) is -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 1

Write the equation of a circle from a situation • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

A sprinkler at (3, −2) waters everything within 4 meters.

Write an equation of the boundary of the watered region in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q1/5

1205

QUESTION ID
S33-F1-Q1

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 + (y + 2)^2 = 16$

GRADING NOTE Form required: the standard form of a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q1**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cell tower at the origin reaches 10 km. Is a phone at (6, 7) within range? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q2/5

1206

QUESTION ID
S33-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes.
$6^2 + 7^2 = 85 < 100$, so it is about 9.2 km away.

GRADING NOTE Any equivalent explanation using distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q2/5

Which equation describes all the points 5 units from (0, 0)?

A. $x^2 - y^2 = 25$

B. $x^2 + y^2 = 25$

C. $x + y = 5$

D. $x^2 + y^2 = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q3/5

1207

QUESTION ID
S33-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. $x^2 + y^2 = 25$

GRADING NOTE The radius squared is 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 1)^2 + (y - 1)^2 = 4$ passes through $(3, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q4/5

1208

QUESTION ID
S33-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $2^2 + 0^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The circle $x^2 + y^2 = 9$ passes through (3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q5/5

1209

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9 + 9 = 18, not 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 2

Write and interpret polynomial models • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner.

Write an equation for its volume V.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q1/5

1211

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

V = x(10 − 2x)(8 − 2x)

GRADING NOTE Any equivalent form. The base is (10 − 2x) by (8 − 2x), and the height is x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner and folding up the sides. Its volume is $V = x(10 - 2x)(8 - 2x)$.

Which values of x make sense in the situation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q2/5

1212

QUESTION ID
S33-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$0 < x < 4$

GRADING NOTE Form required: an inequality. Each side length must be positive, and $8 - 2x > 0$ means $x < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the zeros of $V = x(10 - 2x)(8 - 2x)$?

A. 5 and 4 only

B. 0, 10, and 8

C. 0, −5, and −4

D. 0, 5, and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q3/5

1213

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0, 5, and 4

GRADING NOTE Set each factor equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q3/5

True or false?

The graph of $y = x^3 - x$ crosses the x-axis three times.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q4/5

1214

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

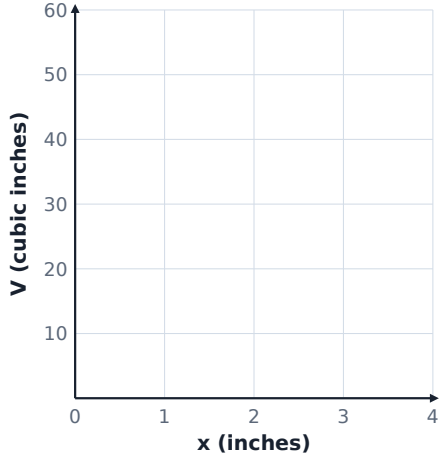
A. True

GRADING NOTE $x(x - 1)(x + 1) = 0$ at $x = -1$, 0, and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q4/5

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner. Its volume is $V = x(10 - 2x)(8 - 2x)$. Graph V for $0 \le x \le 4$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q5/5

1215

QUESTION ID
S33-F2-Q5

TYPE
Drawing / plotting

CORRECT ANSWER
A smooth curve through (0, 0), (1, 48), (2, 48), (3, 24), and (4, 0), highest near x ≈ 1.5 (V ≈ 52.5)

GRADING NOTE Any correct graph through these points with one maximum between x = 1 and x = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q5/5

Represent constraints with equations and inequalities and decide whether solutions are viable

MAIN	Algebra I main question	**HSA-CED.A.3**
BACKWARD 1	Test points in a system of inequalities	**HSA-REI.D.12**
BACKWARD 2	Solve a system of linear equations	**HSA-REI.C.6**
FORWARD 1	Use constraints to optimize a design	**HSG-MG.A.3**
FORWARD 2	Use expected values to compare decisions — nearest related	**HSS-MD.B.7**

SET 34 • HSA-CED.A.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A bakery makes m muffins and b loaves. A muffin uses 1 cup of flour and a loaf uses 3 cups; there are 60 cups. A muffin takes 2 minutes and a loaf takes 5 minutes; there are 140 minutes.

Write the system of constraints.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q1/5

1218

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

m + 3b ≤ 60, 2m + 5b ≤ 140, m ≥ 0, b ≥ 0 (m and b whole numbers)

GRADING NOTE Form required: both resource inequalities; the conditions m ≥ 0 and b ≥ 0 (whole numbers) are also expected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bakery makes m muffins and b loaves. A muffin uses 1 cup of flour and a loaf uses 3 cups; there are 60 cups. A muffin takes 2 minutes and a loaf takes 5 minutes; there are 140 minutes.

Is making 40 muffins and 10 loaves possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q2/5

1219

QUESTION ID

S34-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. It needs 40 + 30 = 70 cups of flour, more than 60 (the 130 minutes would be enough).

GRADING NOTE Any equivalent explanation that checks the constraints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A club needs at least $500 from selling shirts at $15 and hats at $10, and it can sell at most 45 items.

Write the constraints for s shirts and h hats.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q3/5

1220

QUESTION ID

S34-M-Q3

TYPE

Short answer

CORRECT ANSWER

15s + 10h ≥ 500, s + h ≤ 45, s ≥ 0, h ≥ 0 (whole numbers)

GRADING NOTE Form required: both inequalities; s, h ≥ 0 (whole numbers) are also expected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q3/5

A club sells shirts for $15 and hats for $10. It needs at least $500 and can sell at most 45 items.

Which combination meets both constraints?

A. 15 shirts and 25 hats

B. 25 shirts and 15 hats

C. 30 shirts and 20 hats

D. 20 shirts and 10 hats

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q4/5

1221

QUESTION ID

S34-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 25 shirts and 15 hats

GRADING NOTE 375 + 150 = 525 ≥ 500 and 40 ≤ 45. 30 shirts and 20 hats is 50 items; 15 shirts and 25 hats brings $475; 20 shirts and 10 hats brings $400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q4/5

True or false?

If x counts people, x = 12.5 can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q5/5

1222

QUESTION ID

S34-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A number of people must be a whole number, so 12.5 is not viable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q5/5

SET 34 • HSA-CED.A.3

BACKWARD 1

Test points in a system of inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Is (10, 10) a solution of $x + 3y \leq 60$ and $2x + 5y \leq 140$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q1/5

1224

QUESTION ID
S34-B1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. 40 ≤ 60 and 70 ≤ 140.

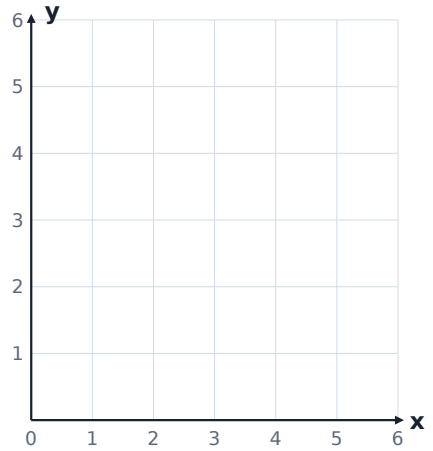
GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph the system $x + y \leq 5$, $x \geq 0$, and $y \geq 0$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q2/5

1225

QUESTION ID

S34-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices (0, 0), (5, 0), and (0, 5), shaded, with solid sides

GRADING NOTE Any correct graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q2/5

Which point is in the solution region of $y \geq 2x$ and $y \leq 6$?

A. (4, 1)

B. (1, 7)

C. (3, 5)

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q3/5

1226

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4)

GRADING NOTE $4 \geq 2$ and $4 \leq 6$. (3, 5) fails $5 \geq 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q3/5

True or false?

The point (0, 0) satisfies $x \geq 0$ and $y \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q4/5

1227

QUESTION ID

S34-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0 \geq 0$ is true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The point (2, 5) satisfies $x + y < 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q5/5

1228

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2 + 5 = 7, which is not less than 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSA-CED.A.3

BACKWARD 2

Solve a system of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + y = 45$

$15x + 10y = 500$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q1/5

1230

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(10, 35)

GRADING NOTE Any equivalent form. $15x + 450 - 10x = 500$, so $x = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$2x + y = 12$

$x + y = 8$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q2/5

1231

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 4)

GRADING NOTE Any equivalent form. Subtract: x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + 3y = 60$

$x + y = 40$

A. (30, 10)

B. (15, 15)

C. (10, 30)

D. (20, 20)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q3/5

1232

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (30, 10)

GRADING NOTE Subtract: 2y = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(10, 35) is a solution of x + y = 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q4/5

1233

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10 + 35 = 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The system x + y = 10 and x + y = 12 has one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q5/5

1234

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No pair can have both sums: no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSA-CED.A.3

FORWARD 1

Use constraints to optimize a design • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A feasible region has vertices (0, 0), (60, 0), and (0, 20).

Find the greatest value of P = 2m + 7b on the region and where it occurs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q1/5

1236

QUESTION ID

S34-F1-Q1

TYPE

Short answer

CORRECT ANSWER

140, at (0, 20)

GRADING NOTE Form required: the value and the point. P = 0, 120, and 140 at the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular pen uses at most 40 m of fence.

What dimensions give the greatest area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q2/5

1237

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 m by 10 m (area 100 m²)

GRADING NOTE Form required: the dimensions. A = w(20 − w) is greatest at w = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q2/5

The cost C = 3x + 2y is checked at the vertices (0, 6), (2, 2), and (6, 0) of a region.

Where is the cost least?

A. (2, 2)

B. (0, 0)

C. (6, 0)

D. (0, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q3/5

1238

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 2)

GRADING NOTE C = 12, 10, and 18 at the vertices. (0, 0) is not in the region.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q3/5

True or false?

For a linear expression on a polygonal region, the greatest value occurs at a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q4/5

1239

QUESTION ID

S34-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the basis of linear programming.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with area 36, a square has the smallest perimeter, so its side is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q5/5

1240

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The side is $\sqrt{36} = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q5/5

FORWARD 2

Use expected values to compare decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Choosing viable options under constraints extends to choosing between options by expected value.

ANSWER KEY Reduced copy of the question page, with its answer

Plan A earns $100 with probability 0.6 and $0 otherwise.
Plan B earns $70 for sure.

Which plan has the greater expected value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q1/5 • NEAREST RELATED

1242

QUESTION ID
S34-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Plan B: $70 compared with $60 for Plan A.

GRADING NOTE Form required: the plan and both expected values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Guessing on a 4-choice question earns 1 point if right and loses 0.25 point if wrong.

What is the expected score from guessing?

Expected score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q2/5 • NEAREST RELATED

1243

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.0625 point

GRADING NOTE Any equivalent form, such as $\frac{1}{16}$. $0.25(1) + 0.75(-0.25) = 0.0625$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q2/5 • NEAREST RELATED

A $5 warranty pays $100 if the item breaks (probability 0.04) and $0 otherwise.

What is the expected net value of buying it?

A. −$5

B. $1

C. $4

D. −$1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q3/5 • NEAREST RELATED

1244

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −$1

GRADING NOTE 100(0.04) − 5 = −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A choice with the higher expected value can still lose money on a single trial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q4/5 • NEAREST RELATED

1245

QUESTION ID
S34-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Expected value describes the long-run average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If a game has an expected value of 0, a player cannot lose money playing it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S34-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q5/5 • NEAREST RELATED

1246

QUESTION ID
S34-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Individual plays can lose; only the long-run average is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S34-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 35 • HSF-IF.A.1 • INTERPRETING FUNCTIONS

Understand a function as a rule that assigns exactly one output to each input, with a domain and a range

MAIN	Algebra I main question	**HSF-IF.A.1**
BACKWARD 1	Decide whether a relation is a function	**8.F.A.1**
BACKWARD 2	Check whether points are on the graph of an equation	**HSA-REI.D.10**
FORWARD 1	Describe transformations of the plane as functions of points	**HSG-CO.A.2**
FORWARD 2	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**

SET 35 • HSF-IF.A.1

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Does the relation {(−2, 4), (0, 1), (2, 4), (3, −1)} define y as a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q1/5

1249

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Each input has exactly one output. (Two inputs may share the output 4.)

GRADING NOTE Form required: yes or no with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain and the range of $f(x) = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q2/5

1250

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

Domain: all real numbers. Range: $y \geq 1$.

GRADING NOTE Form required: both the domain and the range. $x^2 \geq 0$, so $x^2 + 1 \geq 1$, and every value 1 or more is reached.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation does NOT define y as a function of x?

A. $y = x^2$

B. $x = y^2$

C. $y = 2^x$

D. $y = |x|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q3/5

1251

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $x = y^2$

GRADING NOTE For $x = y^2$, the input $x = 4$ has two outputs, $y = 2$ and $y = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = √(x − 3). Over the real numbers, what is the domain of f?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q4/5

1252

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

x ≥ 3

GRADING NOTE Form required: an inequality (or interval [3, ∞)). The expression under the square root must not be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q4/5

True or false?

If f is a function and f(2) = 5, then f(2) can also equal 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q5/5

1253

QUESTION ID
S35-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A function assigns exactly one output to each input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 1

Decide whether a relation is a function • Grade 8 • Functions • 8.F.A.1

A table pairs x = 1, 2, 3, 2 with y = 4, 5, 6, 7. Is y a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q1/5

1255

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

No. The input 2 has two outputs, 5 and 7.

GRADING NOTE Form required: yes or no with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Each button on a vending machine gives exactly one kind of snack. Is the snack a function of the button? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q2/5

1256

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Each input (button) gives exactly one output (snack).

GRADING NOTE Form required: yes or no with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q2/5

Which set of ordered pairs is a function?

A. {(5, 1), (5, 2), (6, 3)}

B. {(0, 0), (0, 1)}

C. {(1, 3), (2, 3), (3, 3)}

D. {(1, 2), (1, 3), (2, 4)}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q3/5

1257

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. {(1, 3), (2, 3), (3, 3)}

GRADING NOTE In {(1, 3), (2, 3), (3, 3)} every input appears once. Each of the other sets repeats an input with a different output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q3/5

True or false?

Two different inputs of a function can have the same output.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q4/5

1258

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, $f(x) = x^2$ gives $f(2) = f(-2) = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A function can pair the input 4 with two different outputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q5/5

1259

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each input has exactly one output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 2

Check whether points are on the graph of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (−2, 5) on the graph of $y = x^2 + 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q1/5

1261

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. $(-2)^2 + 1 = 5$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (a, 10) is on the graph of y = 2x + 4.

Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q2/5

1262

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 2a + 4 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the graph of $y = \sqrt{x}$?

A. (9, −3)

B. (−9, 3)

C. (9, 3)

D. (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q3/5

1263

QUESTION ID

S35-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (9, 3)

GRADING NOTE $\sqrt{9} = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of y = 4 is the set of all points with y-coordinate 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q4/5

1264

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every such point makes the equation true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(1, 3) is on the graph of $y = x^2 + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q5/5

1265

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1^2 + 3 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSF-IF.A.1

FORWARD 1

Describe transformations of the plane as functions of points • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2

The translation T maps (x, y) to (x + 3, y − 2).

Find T(4, 1).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q1/5

1267

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, −1)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A reflection across the y-axis maps (x, y) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q2/5

1268

QUESTION ID
S35-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(−x, y)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q2/5

Which transformation does NOT preserve distance?

A. (x, y) → (x + 5, y)

B. (x, y) → (−x, y)

C. (x, y) → (y, x)

D. (x, y) → (2x, y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q3/5

1269

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (2x, y)

GRADING NOTE (x, y) → (2x, y) stretches horizontal distances. (x, y) → (x + 5, y) is a translation, and (x, y) → (−x, y) and (x, y) → (y, x) are reflections, which keep distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q3/5

True or false?

A transformation of the plane is a function: each input point has exactly one image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q4/5

1270

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is what makes it a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q4/5

True or false?

A rotation of 90° counterclockwise about the origin maps (1, 0) to (0, −1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q5/5

1271

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It maps (1, 0) to (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

FORWARD 2

Restrict a domain to make a function invertible • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $f(x) = x^2$, with all real numbers as its domain, has no inverse function.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q1/5

1273

QUESTION ID
S35-F2-Q1

TYPE
Short answer

CORRECT ANSWER
f(2) = f(−2) = 4, so an inverse would have to send 4 to two different outputs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 1)^2$. Restrict f to the largest interval that contains $x = 3$ on which f is one-to-one, and write the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q2/5

1274

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \geq 1$; $f^{-1}(x) = 1 + \sqrt{x}$, for $x \geq 0$

GRADING NOTE Form required: the interval and the inverse. On $x \geq 1$, f is increasing, and solving $x = (y - 1)^2$ with $y \geq 1$ gives $y = 1 + \sqrt{x}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q2/5

Which function is one-to-one on all real numbers?

A. $f(x) = x^2 - 4$

B. $f(x) = x^3$

C. $f(x) = |x|$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q3/5

1275

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $f(x) = x^3$

GRADING NOTE x^3 takes each output once. x^2, $|x|$, and $x^2 - 4$ give the same output for x and $-x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ with domain $x \geq 0$ has the inverse $f^{-1}(x) = \sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q4/5

1276

QUESTION ID

S35-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE On $x \geq 0$, $\sqrt{(x^2)} = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q4/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q5/5

1277

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only one-to-one functions do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-IF.A.2 • INTERPRETING FUNCTIONS

Use function notation, evaluate functions, and interpret statements in function notation in context

MAIN	Algebra I main question	**HSF-IF.A.2**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve linear equations	**HSA-REI.B.3**
FORWARD 1	Evaluate compositions of functions	**HSF-BF.A.1.c**
FORWARD 2	Read values of an inverse function	**HSF-BF.B.4.c**

SET 36 • HSF-IF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x^2 - 3x + 1.$

Find $f(-2)$.

$f(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q1/5

1280

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 2(4) + 6 + 1 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 64t$ gives a ball's height in feet t seconds after it is hit.

Find h(1), and explain what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q2/5

1281

QUESTION ID
S36-M-Q2

TYPE
Short answer

CORRECT ANSWER
h(1) = 48: one second after it is hit, the ball is 48 feet high.

GRADING NOTE Form required: the value and its meaning (with units).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 5x - 7$.

Find the value of x for which $g(x) = 18$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q3/5

1282

QUESTION ID

S36-M-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $5x - 7 = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

C(n) is the cost in dollars of n tickets.

What does C(4) = 60 mean?

A. 4 tickets cost $60.

B. C equals 4 times 60.

C. Each ticket costs $60.

D. 60 tickets cost $4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q4/5

1283

QUESTION ID

S36-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4 tickets cost $60.

GRADING NOTE The input is the number of tickets and the output is the cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $f(x) = x^2$, then $f(a + 1) = a^2 + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q5/5

1284

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(a + 1) = (a + 1)^2 = a^2 + 2a + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-IF.A.2

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $2x^2 - 3x + 1$ when $x = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q1/5

1286

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $18 - 9 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-16t^2 + 64t$ when $t = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q2/5

1287

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

64

GRADING NOTE Any equivalent form. $-64 + 128$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $3(a - 4)^2$ when $a = 6$.

A. 144

B. 12

C. 6

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q3/5

1288

QUESTION ID
S36-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 12

GRADING NOTE $3(2)^2 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When $x = -1$, $x^2 - x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q4/5

1289

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1 + 1 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When $n = 5$, $2n^2 = 100$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q5/5

1290

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2(25) = 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSF-IF.A.2

BACKWARD 2

Solve linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 5x − 7 = 18.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q1/5

1292

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3(x + 2) = 2x − 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q2/5

1293

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−10

GRADING NOTE Any equivalent form. 3x + 6 = 2x − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B2-Q2**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 0.5x + 3 = 11.

A. x = 7

B. x = 28

C. x = 16

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q3/5

1294

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 16

GRADING NOTE 0.5x = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = -2$ is the solution of $4x + 9 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q4/5

1295

QUESTION ID
S36-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $-8 + 9 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q4/5

True or false?

The solution of $7 - x = 10$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q5/5

1296

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-x = 3$, so $x = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 1

Evaluate compositions of functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x + 4$ and $g(x) = 3x$.

Find $f(g(2))$.

$f(g(2)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q1/5

1298

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. g(2) = 6 and f(6) = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = x - 3$.

Write $f(g(x))$.

$f(g(x)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q2/5

1299

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2$

GRADING NOTE Any equivalent form, such as $x^2 - 6x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x$ and $g(x) = x + 1$.

Find $g(f(5))$.

A. 11

B. 10

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q3/5

1300

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 11

GRADING NOTE $f(5) = 10$ and $g(10) = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q3/5

True or false?

If $f(x) = x - 1$ and $g(x) = x + 1$, then $f(g(x)) = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q4/5

1301

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + 1) - 1 = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $f(x) = 2x$ and $g(x) = x^2$, then $f(g(3)) = 36$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q5/5

1302

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $g(3) = 9$ and $f(9) = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 2

Read values of an inverse function • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

The function f has domain {1, 2, 3, 4}, with f(1) = 3, f(2) = 5, f(3) = 7, and f(4) = 9. Its outputs are all different, so f has an inverse.

Find $f^{-1}(7)$.

$f^{-1}(7)$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q1/5

1304

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. f(3) = 7, so $f^{-1}(7) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x + 1$ is invertible.

Find $f^{-1}(11)$.

$f^{-1}(11) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q2/5

1305

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $2x + 1 = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

f is a one-to-one function with f(4) = 10.

Which statement is true?

A. $f^{-1}(10) = 4$

B. $f^{-1}(4) = 10$

C. $f(10) = 4$

D. $f^{-1}(10) = \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q3/5

1306

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $f^{-1}(10) = 4$

GRADING NOTE The inverse reverses the pair (4, 10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q3/5

True or false?

If f is a one-to-one function, the graph of f^{-1} is the reflection of the graph of f across the line y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q4/5

1307

QUESTION ID
S36-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each point (a, b) on f becomes (b, a) on f^{-1}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f^{-1}(x)$ means $1 \div f(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q5/5

1308

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f^{-1} is the inverse function, not the reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q5/5

SET 37 • HSF-IF.A.3 • INTERPRETING FUNCTIONS

Recognize that sequences are functions whose domain is a set of integers

MAIN	Algebra I main question	**HSF-IF.A.3**
BACKWARD 1	Evaluate functions in function notation	**HSF-IF.A.2**
BACKWARD 2	Generate two numerical patterns and relate their terms	**5.OA.B.3**
FORWARD 1	Repeat a dilation and describe the lengths — nearest related	**HSG-SRT.A.1**
FORWARD 2	Find the sum of a finite geometric series	**HSA-SSE.B.4**

SET 37 • HSF-IF.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A sequence is defined by $f(1) = 3$ and $f(n) = f(n - 1) + 4$ for $n \geq 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q1/5

1311

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

19

GRADING NOTE Any equivalent form. 3, 7, 11, 15, 19.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sequence is defined by $f(1) = 2$, $f(2) = 3$, and $f(n) = f(n - 1) + 2f(n - 2)$ for $n \geq 3$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q2/5

1312

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

27

GRADING NOTE Any equivalent form. $f(3) = 3 + 2(2) = 7$, $f(4) = 7 + 2(3) = 13$, $f(5) = 13 + 2(7) = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula for the sequence 5, 10, 20, 40, ..., with f(1) = 5.

f(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q3/5

1313

QUESTION ID

S37-M-Q3

TYPE

Short answer

CORRECT ANSWER

$5 \cdot 2^{n-1}$

GRADING NOTE Any equivalent form, such as $2.5 \cdot 2^{n}$. Each term is 2 times the one before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q3/5

A sequence f(n) starts with f(1).

What is its domain?

A. The positive integers 1, 2, 3, ...

B. The numbers from 0 to 1

C. All real numbers

D. All integers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q4/5

1314

QUESTION ID

S37-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The positive integers 1, 2, 3, ...

GRADING NOTE A sequence is defined only at the term numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q4/5

True or false?

The graph of the sequence f(n) = 2n + 1 for n = 1, 2, 3, ... is a solid line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q5/5

1315

QUESTION ID

S37-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The domain is the positive integers, so the graph is separate points: (1, 3), (2, 5), (3, 7), ...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 1

Evaluate functions in function notation • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 3n - 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q1/5

1317

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 2^n$.

Find f(5).

f(5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q2/5

1318

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = n^2 + 1$.

Find f(3).

A. 7

B. 10

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q3/5

1319

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

GRADING NOTE $9 + 1 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q3/5

True or false?

If f(n) = 10 − n, then f(10) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q4/5

1320

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10 − 10 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q4/5

True or false?

If f(n) = 4n, then f(3) = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q5/5

1321

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 \cdot 3 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 2

Generate two numerical patterns and relate their terms • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 6.

Write the first 5 terms of each pattern.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q1/5

1323

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A: 0, 3, 6, 9, 12; B: 0, 6, 12, 18, 24

GRADING NOTE Form required: both patterns in order. Each B term is 2 times the matching A term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8.

Write ordered pairs (A term, B term) for the first 4 terms. What relationship do you see?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q2/5

1324

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 0), (4, 8), (8, 16), (12, 24); each B term is 2 times the matching A term.

GRADING NOTE Form required: the four pairs and the relationship.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6.

How do their third terms compare?

A. A's is 3 times B's.

B. B's is 4 more than A's.

C. B's is 3 times A's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q3/5

1325

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. B's is 3 times A's.

GRADING NOTE A: 0, 2, 4. B: 0, 6, 12. 12 = 3 × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q3/5

True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10. Each term of B is twice the matching term of A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q4/5

1326

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A: 0, 5, 10, 15. B: 0, 10, 20, 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q4/5

True or false?

Pattern A starts at 1 and adds 2. Pattern B starts at 2 and adds 4. Each term of B is 4 times the matching term of A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q5/5

1327

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A: 1, 3, 5, 7. B: 2, 6, 10, 14. Each B term is 2 times the matching A term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-IF.A.3

FORWARD 1

Repeat a dilation and describe the lengths • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: Repeated dilations produce a geometric sequence of lengths, a function defined on the positive integers.

ANSWER KEY Reduced copy of the question page, with its answer

A segment of length 3 is dilated by scale factor 2, then the image is dilated by 2 again, and so on.

Write the length after n dilations.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q1/5 • NEAREST RELATED

1329

QUESTION ID
S37-F1-Q1

TYPE
Short answer

CORRECT ANSWER
$3 \cdot 2^n$

GRADING NOTE Any equivalent form. Each dilation multiplies the length by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A dilation with center (0, 0) and scale factor 3 maps (2, −1) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q2/5 • NEAREST RELATED

1330

QUESTION ID
S37-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(6, −3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q2/5 • NEAREST RELATED

A dilation with scale factor $\frac{1}{2}$ maps a segment of length 10 to a segment of what length?

A. 20

B. 9.5

C. 10

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q3/5 • NEAREST RELATED

1331

QUESTION ID
S37-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 5

GRADING NOTE $10 \cdot \frac{1}{2} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q4/5 • NEAREST RELATED

1332

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is a property of dilations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A dilation with scale factor 2 doubles the area of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q5/5 • NEAREST RELATED

1333

QUESTION ID
S37-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Areas are multiplied by $2^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 37 • HSF-IF.A.3

FORWARD 2

Find the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum of the first 5 terms of 2, 6, 18, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q1/5

1335

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

242

GRADING NOTE Any equivalent form. $2(3^5 - 1) \div (3 - 1) = 242$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a formula for the sum of the first n terms of 5, 10, 20, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q2/5

1336

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$5(2^n - 1)$

GRADING NOTE Any equivalent form. a = 5 and r = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 1 + 4 + 16 + 64?

A. 256

B. 85

C. 84

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q3/5

1337

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 85

GRADING NOTE $(4^4 - 1) \div 3 = 85$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q3/5

True or false?

The sum of the first 4 terms of 1, 2, 4, 8, ... is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q4/5

1338

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1 + 2 + 4 + 8 = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q4/5

True or false?

A geometric series with first term 4, ratio 1, and 5 terms has sum 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q5/5

1339

QUESTION ID
S37-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $4 + 4 + 4 + 4 + 4 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret key features of graphs and tables of functions in context

MAIN	Algebra I main question	**HSF-IF.B.4**
BACKWARD 1	Describe a function qualitatively from a situation	**8.F.B.5**
BACKWARD 2	Use vertex form to find a maximum or minimum	**HSA-SSE.B.3.b**
FORWARD 1	Identify key features of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Identify amplitude, midline, and period of a periodic model	**HSF-TF.B.5**

SET 38 • HSF-IF.B.4

MAIN

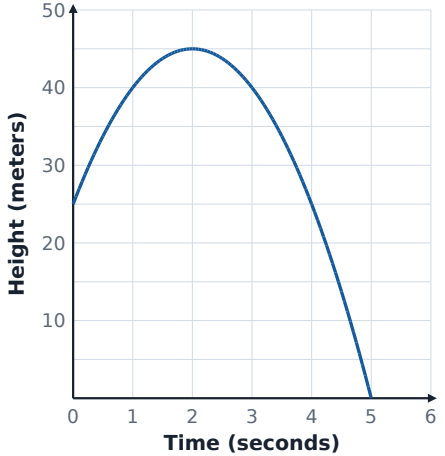
Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

What is the maximum height, and when does it occur?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q1/5

1342

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

45 meters, at t = 2 seconds

GRADING NOTE Form required: the height and the time, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

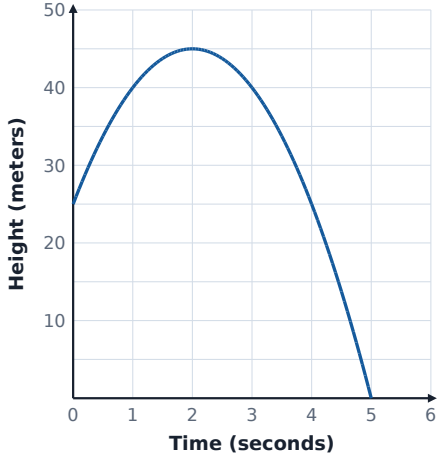
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony (t = 0 is the moment it is thrown).

What is the h-intercept of the graph, and what does it mean?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q2/5

1343

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

25: the ball was thrown from 25 meters above the ground.

GRADING NOTE Form required: the value and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

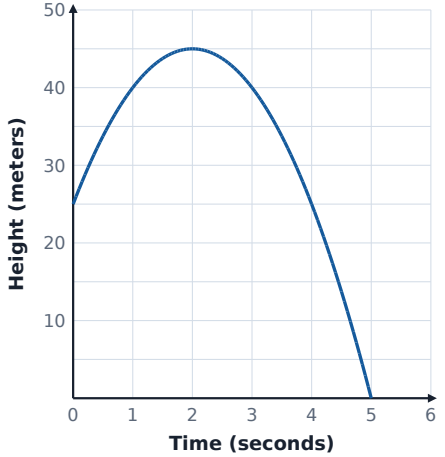
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

When does the ball reach the ground? Over what interval is its height increasing?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q3/5

1344

QUESTION ID

S38-M-Q3

TYPE

Short answer

CORRECT ANSWER

At t = 5 seconds; the height increases for 0 < t < 2.

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q3/5

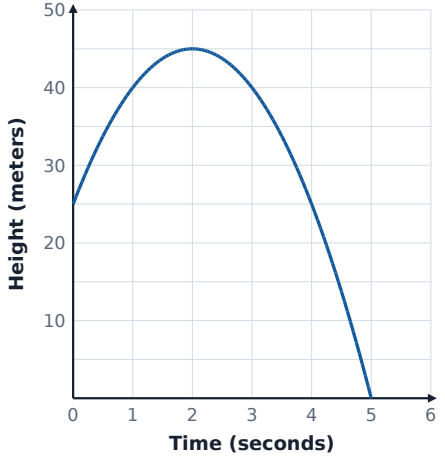
ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

Over what interval is the ball's height decreasing?

A. $0 < t < 2$ **B.** $0 < t < 5$

C. $2 < t < 5$ **D.** $25 < t < 45$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q4/5

1345

QUESTION ID

S38-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $2 < t < 5$

GRADING NOTE After the highest point at $t = 2$, the height falls until the ball lands at $t = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

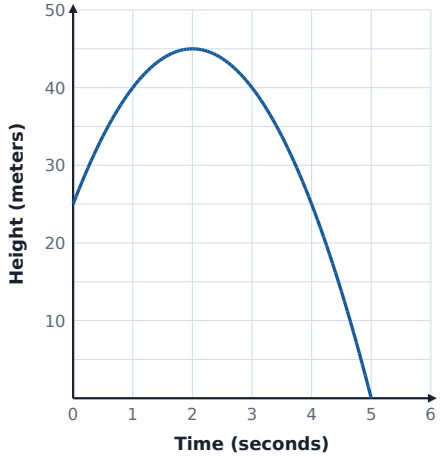
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony. The ball is 25 meters high twice: at t = 0 and at t = 4.

A. True **B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q5/5

1346

QUESTION ID
S38-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The graph is symmetric about t = 2, and 0 and 4 are each 2 seconds from it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 1

Describe a function qualitatively from a situation • Grade 8 • Functions • 8.F.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A car speeds up, drives at a constant speed, then slows to a stop.

Describe the graph of its speed over time.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q1/5

1348

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

It rises, then stays level, then falls to 0.

GRADING NOTE Any equivalent description of the three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q1/5

A graph of water level is horizontal from t = 2 to t = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q2/5

1349

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The water level did not change from t = 2 to t = 5.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which situation has a graph that first increases and then decreases?

A. Savings with a steady deposit each week

B. The height of a ball thrown up that comes back down

C. The distance a parked car travels

D. The height of a burning candle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q3/5

1350

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The height of a ball thrown up that comes back down

GRADING NOTE The ball rises and then falls. A burning candle only gets shorter, a parked car's distance stays the same, and steady deposits only increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

On a distance-time graph, a steeper segment means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q4/5

1351

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE More distance in the same time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A decreasing graph always means the quantity is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q5/5

1352

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A quantity can decrease while staying positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 2

Use vertex form to find a maximum or minimum • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

ANSWER KEY Reduced copy of the question page, with its answer

Write $-5t^2 + 20t + 25$ in the form $a(t - h)^2 + k$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q1/5

1354

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$-5(t - 2)^2 + 45$

GRADING NOTE Form required: $a(t - h)^2 + k$. $-5(t^2 - 4t) + 25 = -5(t - 2)^2 + 20 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the maximum value of $-(x - 3)^2 + 7$, and for what x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q2/5

1355

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7, at x = 3

GRADING NOTE Form required: the value and the x-value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q2/5

What is the minimum value of $2(x + 1)^2 - 8$?

A. -8

B. -1

C. 2

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q3/5

1356

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −8

GRADING NOTE $2(x + 1)^2 \geq 0$, and it is 0 at $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q3/5

True or false?

$-5(t-2)^2 + 45$ is greatest when $t = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q4/5

1357

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The squared term is 0 there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

The minimum value of $(x - 4)^2 + 1$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q5/5

1358

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The minimum value is 1, at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 1

Identify key features of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

Describe the end behavior of $f(x) = -2x^3 + x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q1/5

1360

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$.

GRADING NOTE Any equivalent description. The leading term $-2x^3$ decides the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q1/5

Find the zeros of f(x) = x(x − 2)(x + 3).

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q2/5

1361

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0, 2, and −3

GRADING NOTE Form required: all three zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function rises on the right and falls on the left?

A. $f(x) = -x^4$

B. $f(x) = x^3 - 4x$

C. $f(x) = x^2$

D. $f(x) = -x^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q3/5

1362

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $f(x) = x^3 - 4x$

GRADING NOTE An odd degree with a positive leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A polynomial function of degree 4 has at most 4 real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q4/5

1363

QUESTION ID
S38-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

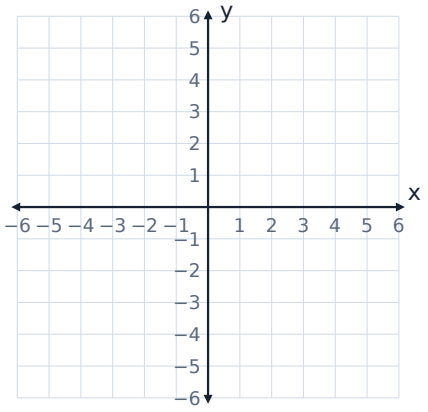
GRADING NOTE A degree-n polynomial has at most n zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = -x^2(x - 3)$. Mark the zeros and show what the graph does at each zero and at the ends. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q5/5

1364

QUESTION ID

S38-F1-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A curve that rises to the left, touches the x-axis at 0 and turns (double zero), rises to a peak between 0 and 3, crosses the x-axis at 3, and falls to the right

GRADING NOTE Any correct graph: touch at 0, cross at 3, end behavior of $-x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 2

Identify amplitude, midline, and period of a periodic model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel seat's height is $h(t) = 30 - 25\cos(\pi t \div 5)$ meters after t minutes.

What are the greatest and least heights?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q1/5

1366

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

55 meters and 5 meters

GRADING NOTE Form required: both heights. cos is between −1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q1/5

What is the period of $h(t) = 30 - 25\cos(\pi t \div 5)$?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q2/5

1367

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 minutes

GRADING NOTE Any equivalent form. $2\pi \div (\pi \div 5) = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q2/5

What is the midline of y = 4sin x + 2?

A. y = 4

B. y = 2

C. y = 6

D. y = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q3/5

1368

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2

GRADING NOTE The graph goes from −2 to 6, centered on y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q3/5

True or false?

The amplitude of $y = 3\sin x$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q4/5

1369

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph goes 3 units above and below its midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q4/5

True or false?

The function y = sin x has period π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q5/5

1370

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The period is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q5/5

SET 78 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Sketch graphs that show key features given in a verbal description

MAIN	Algebra I main question	**HSF-IF.B.4**
BACKWARD 1	Sketch a graph from a description	**8.F.B.5**
BACKWARD 2	Write and graph equations on labeled axes	**HSA-CED.A.2**
FORWARD 1	Sketch polynomial graphs from a description of their features	**HSF-IF.C.7.c**
FORWARD 2	Sketch periodic graphs from a description	**HSF-TF.B.5**

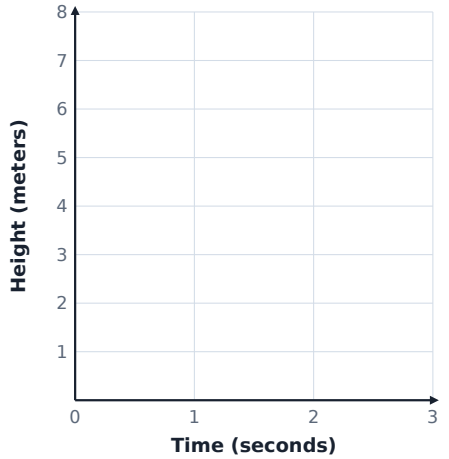
SET 78 • HSF-IF.B.4

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A ball is thrown up from a 2-m platform. It reaches its highest point, 7 m, after 1 second, and it lands on the ground after about 2.2 seconds.

Sketch its height against time. Label the starting point, the maximum, and the landing time.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q1/5

1373

QUESTION ID

S78-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A curve starting at (0, 2), rising to a peak at (1, 7), and falling to the t-axis near (2.2, 0), with those points labeled

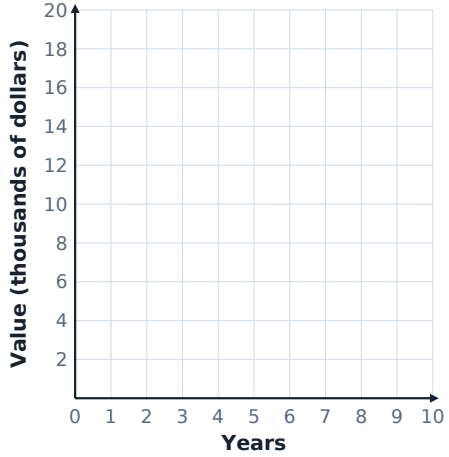
GRADING NOTE Any correct sketch: the intercept 2, the maximum (1, 7), increasing then decreasing, and the landing time near 2.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q1/5

A car is worth $20,000 when new, and it loses 15% of its value each year, never reaching $0.

Sketch its value for 0 to 10 years. Show the starting value and what happens as time goes on.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q2/5

1374

QUESTION ID

S78-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A decreasing curve from (0, 20) that falls more slowly each year and stays above 0 (about 4 at 10 years)

GRADING NOTE Any correct sketch: intercept 20, always decreasing, curving toward but never reaching 0. $20(0.85)^{10} \approx 3.9$.

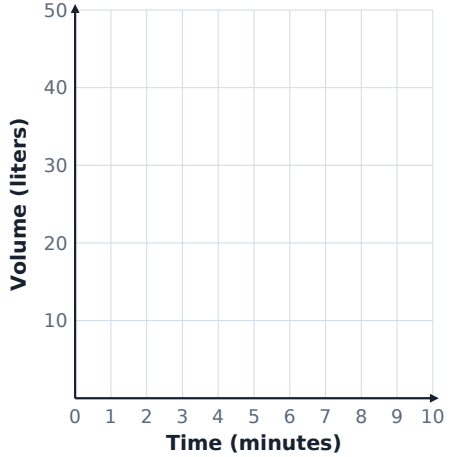
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 50-liter tank drains at a constant rate until it is empty after 5 minutes, stays empty for 2 minutes, and then refills at a constant rate to 30 liters in 3 minutes.

Sketch the volume over the 10 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q3/5

1375

QUESTION ID

S78-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

A segment from (0, 50) to (5, 0), a horizontal segment from (5, 0) to (7, 0), and a segment from (7, 0) to (10, 30)

GRADING NOTE Any correct sketch with the three pieces and their intervals of decrease, constancy, and increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q3/5

A town's population grew slowly at first and then faster and faster.

Which graph shape fits?

A. Increasing and leveling off

B. Increasing and curving upward

C. A horizontal line

D. Decreasing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q4/5

1376

QUESTION ID

S78-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Increasing and curving upward

GRADING NOTE Growth that keeps speeding up gives an increasing graph that gets steeper.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q4/5

True or false?

A graph of the temperature of a cup of coffee left on a table falls at a constant rate until it reaches 0 °C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q5/5

1377

QUESTION ID

S78-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The coffee cools quickly at first and then more slowly, leveling off near room temperature, not 0 °C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q5/5

SET 78 • HSF-IF.B.4

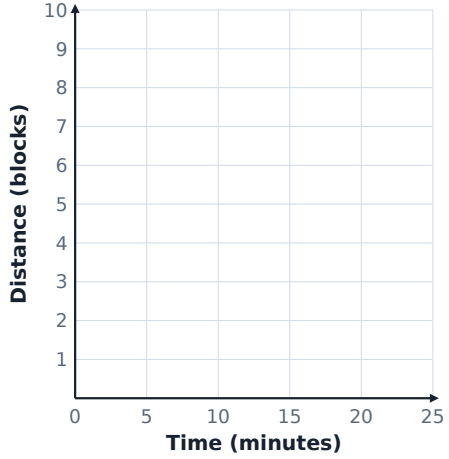
BACKWARD 1

Sketch a graph from a description • Grade 8 • Functions • 8.F.B.5

ANSWER KEY Reduced copy of the question page, with its answer

You walk away from home at a steady pace for 10 minutes, rest for 5 minutes, then walk home faster in 10 minutes.

Sketch your distance from home over 25 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q1/5

1379

GRADING NOTE Any correct qualitative sketch with the three pieces in order.

QUESTION ID

S78-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A rising segment for 0 to 10 minutes, a horizontal segment for 10 to 15, and a steeper falling segment back to 0 at 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph of a plant's height rises steeply and then levels off. Describe what happened to the plant.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q2/5

1380

QUESTION ID

S78-B1-Q2

TYPE

Short answer

CORRECT ANSWER

It grew quickly at first, then its growth slowed until its height stayed about the same.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q2/5

Which situation has a graph that stays constant (horizontal) for a while?

A. A rising balloon's height

B. A burning candle's height

C. A parked car's distance from home

D. A falling ball's height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q3/5

1381

QUESTION ID

S78-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A parked car's distance from home

GRADING NOTE A parked car does not move.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q3/5

True or false?

A graph that rises and then falls has a highest point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q4/5

1382

QUESTION ID

S78-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It changes from increasing to decreasing at a maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q4/5

True or false?

A horizontal segment on a speed-time graph means the car is stopped.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q5/5

1383

QUESTION ID

S78-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means the speed is constant, which may not be 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSF-IF.B.4

BACKWARD 2

Write and graph equations on labeled axes • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation for the area A of a square with side s. What kind of graph does it have for s ≥ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q1/5

1385

QUESTION ID

S78-B2-Q1

TYPE

Short answer

CORRECT ANSWER

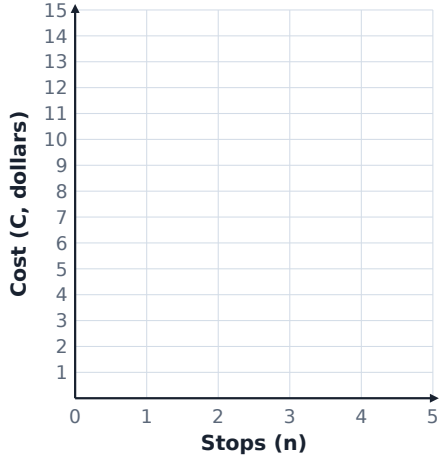
$A = s^2$; half of a parabola, starting at (0, 0) and curving upward

GRADING NOTE Form required: the equation and the shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q1/5

Graph C = 5 + 2n, the cost in dollars of a ride with n stops, for n = 0 to 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q2/5

1386

QUESTION ID

S78-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points (or a line) from (0, 5) to (5, 15), rising by 2 for each stop

GRADING NOTE Any correct graph; separate points are expected because n is a whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A $1,200 laptop loses $200 of value each year. Which equation gives its value V after t years?

A. V = 1,200 − 200t

B. V = 200 − 1,200t

C. V = 1,200 + 200t

D. $V = 1{,}200(0.8)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q3/5

1387

QUESTION ID
S78-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. V = 1,200 − 200t

GRADING NOTE Start at 1,200 and subtract 200 per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q3/5

True or false?

An equation in two variables can be graphed on a coordinate plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q4/5

1388

QUESTION ID

S78-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each solution is a point (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of d = 50t is a curve.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q5/5

1389

QUESTION ID

S78-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a straight line through the origin with slope 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q5/5

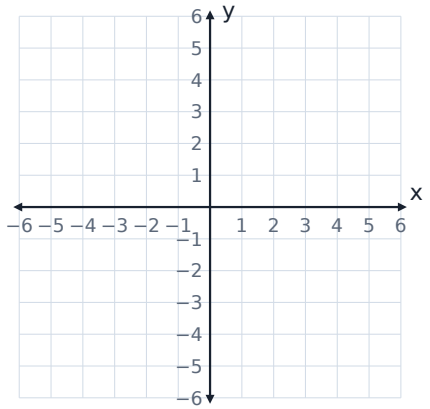
SET 78 • HSF-IF.B.4

FORWARD 1

Sketch polynomial graphs from a description of their features • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

Sketch a cubic function that falls to the left, rises to the right, and has zeros at −2, 1, and 4. (The y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q1/5

1391

QUESTION ID

S78-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −2, 1, and 4, falling to the left and rising to the right, with one turn between −2 and 1 and one between 1 and 4

GRADING NOTE Any correct sketch, for example of y = (x + 2)(x − 1)(x − 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A polynomial graph touches the x-axis at x = 3 and turns around there. What can you say about the factor x − 3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q2/5

1392

QUESTION ID

S78-F1-Q2

TYPE

Short answer

CORRECT ANSWER

It appears an even number of times, for example $(x - 3)^2$.

GRADING NOTE Any equivalent explanation (even multiplicity).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which polynomial function has a graph that falls on both ends?

A. $y = -x^4 + 3x$

B. $y = x^3$

C. $y = -x^3$

D. $y = x^4 - 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q3/5

1393

QUESTION ID

S78-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = -x^4 + 3x$

GRADING NOTE Even degree with a negative leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q3/5

True or false?

The graph of a degree-5 polynomial with a positive leading coefficient rises to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q4/5

1394

QUESTION ID
S78-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Odd degree, positive leading coefficient: falls left, rises right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q4/5

True or false?

The graph of a cubic function can rise on both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q5/5

1395

QUESTION ID
S78-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A cubic rises on one end and falls on the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q5/5

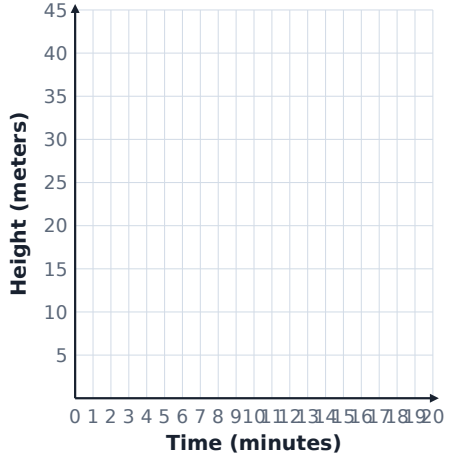
SET 78 • HSF-IF.B.4

FORWARD 2

Sketch periodic graphs from a description • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel seat starts at its lowest height, 2 m, reaches its highest height, 42 m, after 5 minutes, and repeats every 10 minutes.

Sketch its height over 20 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q1/5

1397

QUESTION ID

S78-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A smooth wave starting at (0, 2), up to (5, 42), down to (10, 2), up to (15, 42), down to (20, 2), with midline 22

GRADING NOTE Any correct sketch with the minimum 2, maximum 42, period 10, and starting point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q1/5

Daylight lasts 15 hours in late June and 9 hours in late December. Describe a graph of daylight hours over a year.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q2/5

1398

QUESTION ID

S78-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A wave with maximum 15 (June), minimum 9 (December), midline 12, and period 12 months.

GRADING NOTE Form required: the maximum, minimum, midline, and period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q2/5

Which feature of a periodic graph is the time for one full cycle?

A. The maximum

B. The midline

C. The amplitude

D. The period

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q3/5

1399

QUESTION ID

S78-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The period

GRADING NOTE The period is the length of one cycle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q3/5

True or false?

The midline of a periodic graph is halfway between its maximum and minimum values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q4/5

1400

QUESTION ID
S78-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Midline = (maximum + minimum) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q4/5

True or false?

A tide graph that repeats every 12.4 hours has amplitude 12.4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q5/5

1401

QUESTION ID
S78-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 12.4 hours is the period; the amplitude is a height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q5/5

SET 39 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

Relate the domain of a function to the quantity it describes

MAIN	Algebra I main question	**HSF-IF.B.5**
BACKWARD 1	Decide whether a solution is viable	**HSA-CED.A.3**
BACKWARD 2	Find where a quadratic model equals 0	**HSA-REI.B.4.b**
FORWARD 1	Use trigonometric ratios as functions of acute angles — nearest r..	**HSG-SRT.C.6**
FORWARD 2	Extend trigonometric functions to all real numbers with the unit ci…	**HSF-TF.A.2**

SET 39 • HSF-IF.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

n(k) = 12k gives the number of eggs in k full cartons.
The store has 30 cartons.

What is a reasonable domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q1/5

1404

QUESTION ID
S39-M-Q1

TYPE
Short answer

CORRECT ANSWER
The whole numbers 0, 1, 2, ..., 30

GRADING NOTE Form required: the domain, stated as whole numbers from 0 to 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 64t$ gives a ball's height in feet from the moment it is hit until it lands.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q2/5

1405

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

$0 \le t \le 4$

GRADING NOTE Form required: an inequality (or interval). $h(t) = -16t(t - 4) = 0$ at $t = 0$ and $t = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A taxi fare is C(m) = 3 + 2.5m dollars for m miles, with a limit of 40 miles per ride.

What is the domain?

A. All real numbers

B. m ≥ 3

C. 3 ≤ m ≤ 103

D. 0 ≤ m ≤ 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q3/5

1406

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0 ≤ m ≤ 40

GRADING NOTE The input is miles, from 0 to the 40-mile limit. 3 ≤ C ≤ 103 is the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has perimeter 20. Its area is $A(w) = w(10 - w)$, where w is the width.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q4/5

1407

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

$0 < w < 10$

GRADING NOTE Form required: an inequality. Both w and the length $10 - w$ must be positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q4/5

True or false?

P(n) = 50n gives the number of seats on n buses.
n = 2.5 is in a reasonable domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q5/5

1408

QUESTION ID
S39-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The number of buses must be a whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 1

Decide whether a solution is viable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Solving a problem gives 7.4 buses needed to carry every student.

What is the viable answer? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q1/5

1410

QUESTION ID
S39-B1-Q1

TYPE
Short answer

CORRECT ANSWER
8 buses: the number must be a whole number, and 7 buses are not enough.

GRADING NOTE Form required: the number and a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q1/5

The number of tickets t must satisfy $5t \leq 60$ and $t \geq 0$.

What whole numbers of tickets are viable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q2/5

1411

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0, 1, 2, …, 12

GRADING NOTE Any equivalent form, such as the whole numbers from 0 to 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value is NOT viable for the number of people in a room?

A. 40

B. 0

C. 12

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q3/5

1412

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3

GRADING NOTE A count cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A length of 0.25 meter can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q4/5

1413

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lengths can be fractions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q4/5

True or false?

$w(10 - w) = 24$ has solutions $w = 4$ and $w = 6$, but only one of them is a viable width.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q5/5

1414

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both are viable: width 4 with length 6, or width 6 with length 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 2

Find where a quadratic model equals 0 • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-16t^2 + 64t = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q1/5

1416

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

t = 0 or t = 4

GRADING NOTE Form required: both solutions. $-16t(t - 4) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $w(10 - w) = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q2/5

1417

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$w = 0$ or $w = 10$

GRADING NOTE Form required: both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q2/5

Solve $-16t^2 + 64t + 80 = 0$.

A. $t = 4$ or $t = -2$

B. $t = 5$ or $t = -1$

C. $t = 5$ only

D. $t = -5$ or $t = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q3/5

1418

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $t = 5$ or $t = -1$

GRADING NOTE $t^2 - 4t - 5 = (t - 5)(t + 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q3/5

True or false?

$t = 4$ is a solution of $t^2 - 4t = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S39-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q4/5

1419

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $16 - 16 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S39-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q5/5

1420

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = -9$ has no real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-IF.B.5

FORWARD 1

Use trigonometric ratios as functions of acute angles • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: A trigonometric ratio defined by a right triangle has a domain limited to acute angles, a domain set by the situation.

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C, with AC = 4, BC = 3, and AB = 5.

Find sin A.

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q1/5 • NEAREST RELATED

1422

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE Any equivalent form, such as 0.6. sin A = opposite ÷ hypotenuse = BC ÷ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

For which angle measures θ is sin θ defined as a ratio of sides of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q2/5 • NEAREST RELATED

1423

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0° < θ < 90°

GRADING NOTE Form required: an inequality. Only the acute angles of a right triangle are used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q2/5 • NEAREST RELATED

In a right triangle, $\cos A = \frac{12}{13}$.

What is sin A?

A. $\frac{5}{13}$

B. $\frac{12}{5}$

C. $\frac{1}{13}$

D. $\frac{13}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q3/5 • NEAREST RELATED

1424

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/13

GRADING NOTE The sides are 5, 12, 13, so the opposite side is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In a right triangle, sin A = cos B when A and B are the two acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q4/5 • NEAREST RELATED

1425

QUESTION ID
S39-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The side opposite A is adjacent to B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

For an acute angle θ, sin θ can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q5/5 • NEAREST RELATED

1426

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The opposite side is shorter than the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 39 • HSF-IF.B.5

FORWARD 2

Extend trigonometric functions to all real numbers with the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

Find cos π.

cos π = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q1/5

1428

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. The point at angle π is (−1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q1/5

Find sin(3π ÷ 2).

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q2/5

1429

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. The point is (0, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q2/5

With the unit circle, what is the domain of f(θ) = sin θ?

A. All real numbers

B. $0 < \theta < \pi \div 2$

C. $\theta \geq 0$

D. $0 \leq \theta \leq 2\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q3/5

1430

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. All real numbers

GRADING NOTE Any real number of radians, positive or negative, gives a point on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q3/5

True or false?

A negative radian measure gives a clockwise rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q4/5

1431

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Positive angles turn counterclockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q4/5

True or false?

sin θ is defined only for acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q5/5

1432

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The unit circle defines it for every real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate, interpret, and estimate the average rate of change of a function over an interval

MAIN	Algebra I main question	**HSF-IF.B.6**
BACKWARD 1	Find the rate of change of a linear function	**8.F.B.4**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Build a periodic model from amplitude, period, and midline	**HSF-TF.B.5**
FORWARD 2	Analyze rational functions: rates of change and asymptotes	**HSF-IF.C.7.d**

SET 40 • HSF-IF.B.6

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2 + 2x$.

Find the average rate of change from $x = 1$ to $x = 4$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q1/5

1435

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. $(f(4) - f(1)) \div (4 - 1) = (24 - 3) \div 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A town had 12,000 people in 2010 and 15,600 in 2022.

Find the average rate of change, with units.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q2/5

1436

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

300 people per year

GRADING NOTE Form required: the rate with its units. 3,600 ÷ 12 = 300.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q2/5

The table gives the distance d a cart has traveled after t seconds.

Over which 2-second interval was its average speed greatest? Give that speed.

t (seconds)	0	2	4	6
d (meters)	0	30	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q3/5

1437

QUESTION ID

S40-M-Q3

TYPE

Short answer

CORRECT ANSWER

From t = 2 to t = 4, at 35 meters per second

GRADING NOTE Form required: the interval and the speed. The speeds are 15, 35, and 25 m/s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q3/5

What is the average rate of change of $g(x) = 3^x$ from $x = 0$ to $x = 2$?

A. 4

B. 3

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q4/5

1438

QUESTION ID

S40-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $(9 - 1) \div 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q4/5

True or false?

A linear function has the same average rate of change over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q5/5

1439

QUESTION ID
S40-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its average rate of change is always its slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 1

Find the rate of change of a linear function • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the rate of change of the line through (2, 5) and (6, 17).

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q1/5

1441

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 12 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A tank had 50 liters at t = 0 and 20 liters at t = 10 minutes, draining steadily.

What is the rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q2/5

1442

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−3 liters per minute

GRADING NOTE Any equivalent form, with units. (20 − 50) ÷ 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the rate of change of $y = 7 - 4x$?

A. 7

B. -4

C. $-\frac{4}{7}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q3/5

1443

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −4

GRADING NOTE The coefficient of x is −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A negative rate of change means the quantity is decreasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q4/5

1444

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outputs fall as inputs rise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q4/5

True or false?

The rate of change from (0, 3) to (2, 3) is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q5/5

1445

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (3 − 3) ÷ 2 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A runner goes $\frac{3}{4}$ mile in $\frac{1}{8}$ hour.

What is the speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q1/5

1447

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 miles per hour

GRADING NOTE Any equivalent form. $\frac{3}{4} \div \frac{1}{8} = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q1/5

A car uses 12 gallons to go 300 miles.

How many miles per gallon?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q2/5

1448

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q2/5

A recipe uses $\frac{2}{3}$ cup of sugar for $\frac{1}{3}$ of a batch.

How much sugar is used per batch?

A. $\frac{1}{2}$ cup

B. 1 cup

C. 2 cups

D. $\frac{2}{9}$ cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q3/5

1449

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 cups

GRADING NOTE $\frac{2}{3} \div \frac{1}{3} = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

120 km in 1.5 hours is 80 km per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q4/5

1450

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 120 ÷ 1.5 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$18 for 4 pounds is $4 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q5/5

1451

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 18 ÷ 4 = $4.50 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 1

Build a periodic model from amplitude, period, and midline • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel has diameter 40 m, its center is 25 m above the ground, and it turns once every 8 minutes. A rider starts at the lowest point.

Write a model h(t) = A cos(Bt) + D for the rider's height after t minutes.

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q1/5

1453

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−20cos(πt ÷ 4) + 25

GRADING NOTE Any equivalent form. Amplitude 20, midline 25, B = 2π ÷ 8 = π ÷ 4; A is negative because the ride starts at the lowest point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q1/5

$h(t) = -20\cos(\pi t \div 4) + 25$ gives a rider's height in meters after t minutes.

Give the amplitude, the period, and the midline.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q2/5

1454

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Amplitude 20 m; period 8 minutes; midline h = 25

GRADING NOTE Form required: all three. $2\pi \div (\pi \div 4) = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q2/5

Daily high temperatures in a town vary from a low of 4 °C to a high of 28 °C over a 365-day year.

Which values fit a sinusoidal model?

A. Amplitude 24, midline 16, period 365 days

B. Amplitude 16, midline 12, period 365 days

C. Amplitude 12, midline 14, period 182.5 days

D. Amplitude 12, midline 16, period 365 days

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q3/5

1455

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Amplitude 12, midline 16, period 365 days

GRADING NOTE Amplitude (28 − 4) ÷ 2 = 12; midline (28 + 4) ÷ 2 = 16; one cycle per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q3/5

True or false?

A model $y = A\cos(Bt) + D$ with period 12 hours has $B = \pi \div 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q4/5

1456

QUESTION ID

S40-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2\pi \div B = 12$ gives $B = \pi \div 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q4/5

True or false?

A tide that rises from 1 m to 5 m and falls back again has amplitude 4 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q5/5

1457

QUESTION ID
S40-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The amplitude is half the distance from low to high: (5 − 1) ÷ 2 = 2 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 2

Analyze rational functions: rates of change and asymptotes • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 12 \div x$.

Find the average rate of change from $x = 2$ to $x = 6$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q1/5

1459

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. $(2 - 6) \div (6 - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q1/5

Give the vertical and horizontal asymptotes of $f(x) = 1 \div (x - 3) + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q2/5

1460

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Vertical x = 3; horizontal y = 2

GRADING NOTE Form required: both asymptotes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q2/5

Where is f(x) = (x + 1) ÷ (x − 4) undefined?

A. x = −4

B. x = 1

C. x = −1

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q3/5

1461

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 4

GRADING NOTE The denominator is 0 at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q3/5

True or false?

The graph of $y = 1 \div x$ gets closer to the x-axis as x gets large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q4/5

1462

QUESTION ID
S40-F2-Q4

TYPE
True or false

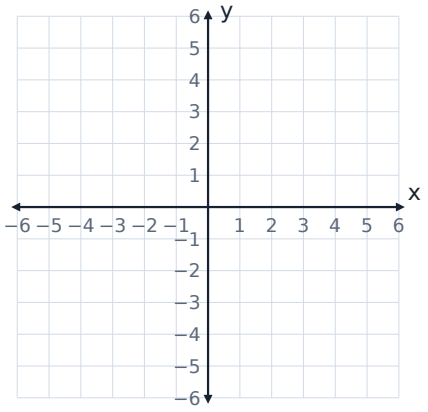
CORRECT ANSWER
A. True

GRADING NOTE $1 \div x$ approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q4/5

Graph f(x) = 1 ÷ (x − 2) + 1. Draw both asymptotes as lines and mark the intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q5/5

1463

QUESTION ID

S40-F2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Vertical asymptote x = 2 and horizontal asymptote y = 1; the curve passes through the x-intercept (1, 0) and the y-intercept (0, 0.5), with one branch below and left of (2, 1) and one above and right

GRADING NOTE Any correct graph. 1 ÷ (x − 2) = −1 gives x = 1; f(0) = −0.5 + 1 = 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q5/5

SET 79 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Estimate the average rate of change of a function from its graph

MAIN	Algebra I main question	**HSF-IF.B.6**
BACKWARD 1	Find a rate of change from two points	**8.F.B.4**
BACKWARD 2	Read a unit rate from a graph	**7.RP.A.2.b**
FORWARD 1	Estimate average rates of change from a polynomial graph	**HSF-IF.C.7.c**
FORWARD 2	Estimate average rates of change from a rational-function graph	**HSF-IF.C.7.d**

SET 79 • HSF-IF.B.6

MAIN

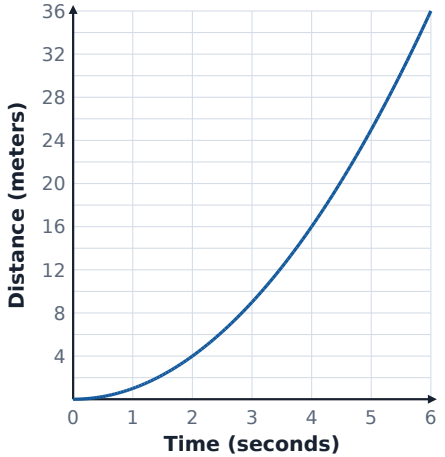
Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows the distance d, in meters, that a cart has rolled after t seconds.

Estimate the average rate of change from t = 1 to t = 3, with units.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q1/5

1466

QUESTION ID
S79-M-Q1

TYPE
Short answer

CORRECT ANSWER
About 4 meters per second

GRADING NOTE Form required: an estimate with units. Accept 3.5 to 4.5 m/s: the graph passes about (1, 1) and (3, 9), and (9 − 1) ÷ 2 = 4.

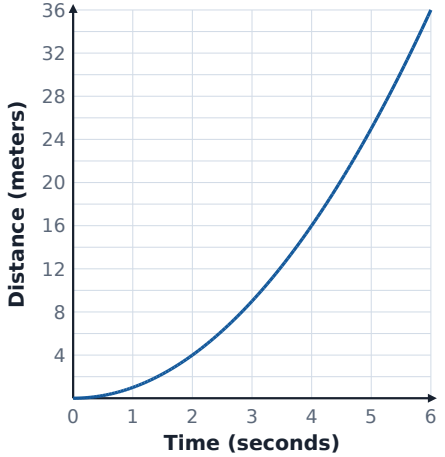
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q1/5

The graph shows the distance d, in meters, that a cart has rolled after t seconds.

Estimate the average rate of change from t = 3 to t = 5. Is the cart speeding up?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q2/5

1467

QUESTION ID

S79-M-Q2

TYPE

Short answer

CORRECT ANSWER

About 8 meters per second; yes, the rate is greater than on earlier intervals.

GRADING NOTE Form required: an estimate (7 to 9 m/s) and the conclusion. (25 − 9) ÷ 2 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

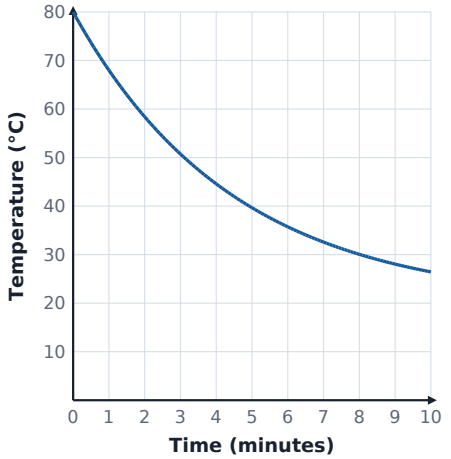
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows the temperature of a cup of coffee.

Estimate the average rate of change of the temperature over the first 5 minutes, with units.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q3/5

1468

QUESTION ID

S79-M-Q3

TYPE

Short answer

CORRECT ANSWER

About −8 °C per minute

GRADING NOTE Form required: an estimate with units. Accept −7 to −9 °C per minute: from about 80 °C to about 40 °C in 5 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q3/5

The graph shows the temperature of a cup of coffee.

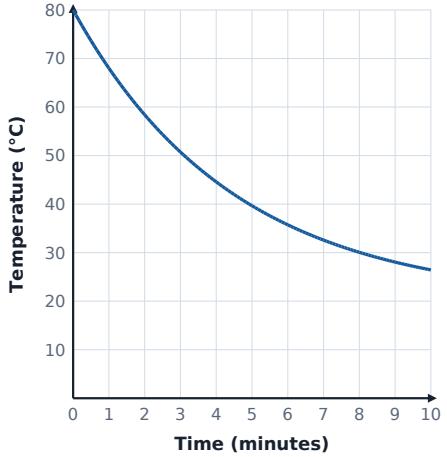
Over which 2-minute interval is the coffee cooling fastest on average?

A. 4 to 6 minutes

B. 8 to 10 minutes

C. 0 to 2 minutes

D. 6 to 8 minutes



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q4/5

1469

QUESTION ID

S79-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0 to 2 minutes

GRADING NOTE The graph is steepest at the start; it flattens as the coffee nears room temperature.

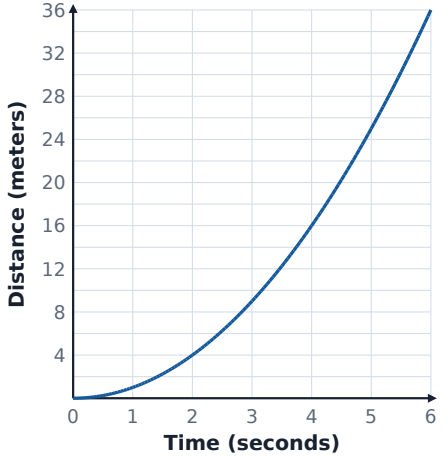
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q4/5

True or false?

The average rate of change of d from t = 0 to t = 6 equals the slope of the segment joining (0, 0) and (6, 36) on the graph.

A. True **B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q5/5

1470

QUESTION ID
S79-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The average rate of change is the slope of the secant line: 36 ÷ 6 = 6 m/s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q5/5

SET 79 • HSF-IF.B.6

BACKWARD 1

Find a rate of change from two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (2, 10) and (6, 30). Find its rate of change.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q1/5

1472

QUESTION ID

S79-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 20 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q1/5

A straight-line graph of a hike shows 3 km at 1 hour and 9 km at 3 hours. What is the rate of change, with units?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q2/5

1473

QUESTION ID

S79-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3 km per hour

GRADING NOTE Any equivalent form. 6 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q2/5

What is the slope of the line through (0, 50) and (5, 25)?

A. 5

B. −5

C. 0.2

D. −25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q3/5

1474

QUESTION ID

S79-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −5

GRADING NOTE $(25 - 50) \div 5 = -5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q3/5

True or false?

The slope between two points on a line is (change in y) ÷ (change in x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q4/5

1475

QUESTION ID

S79-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition of slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q4/5

True or false?

The line through (1, 4) and (3, 4) has rate of change 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q5/5

1476

QUESTION ID

S79-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(4 - 4) \div 2 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSF-IF.B.6

BACKWARD 2

Read a unit rate from a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

The graph of a proportional relationship passes through (4, 12). What is the unit rate?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q1/5

1478

QUESTION ID

S79-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 12 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q1/5

A proportional cost graph passes through (0, 0) and (5, 7.5). What is the cost per item?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q2/5

1479

QUESTION ID

S79-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$1.50

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A proportional graph of miles against gallons passes through (3, 90). What is the unit rate?

A. 3 miles per gallon

B. 270 miles per gallon

C. 90 miles per gallon

D. 30 miles per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q3/5

1480

QUESTION ID
S79-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 30 miles per gallon

GRADING NOTE 90 ÷ 3 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q3/5

True or false?

On a proportional graph, the point (1, r) shows the unit rate r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q4/5

1481

QUESTION ID

S79-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE At 1 unit of x, y equals the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q4/5

True or false?

A proportional graph passes through (2, 5), so its unit rate is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q5/5

1482

QUESTION ID

S79-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5 \div 2 = 2.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSF-IF.B.6

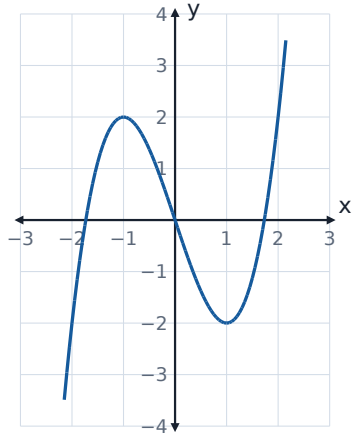
FORWARD 1

Estimate average rates of change from a polynomial graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows $f(x) = x^3 - 3x$.

Estimate the average rate of change from $x = -1$ to $x = 1$.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q1/5

1484

QUESTION ID

S79-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form. $(f(1) - f(-1)) \div 2 = (-2 - 2) \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

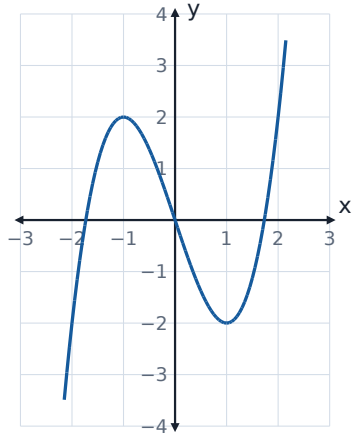
High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows $f(x) = x^3 - 3x$.

Estimate the average rate of change from $x = 1$ to $x = 2$.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q2/5

1485

QUESTION ID

S79-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $(2 - (-2)) \div 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows $f(x) = x^3 - 3x$.

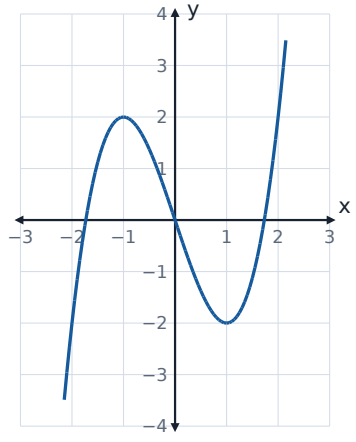
On which interval is f increasing?

A. $-1 < x < 1$

B. $1 < x < 2$

C. $-1 < x < 0$

D. $0 < x < 1$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q3/5

1486

QUESTION ID
S79-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 1 < x < 2

GRADING NOTE f falls between its turning points at $x = -1$ and $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q3/5

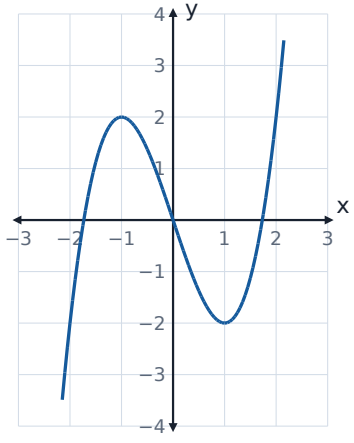
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The average rate of change of $f(x) = x^3 - 3x$ from $x = -2$ to $x = 2$ is 1.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q4/5

1487

QUESTION ID
S79-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $(2 - (-2)) \div 4 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q4/5

True or false?

A positive average rate of change on an interval means the function increases at every point of the interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q5/5

1488

QUESTION ID
S79-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For $f(x) = x^3 - 3x$ on $-2 \le x \le 2$ the average rate is positive, but f decreases between -1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSF-IF.B.6

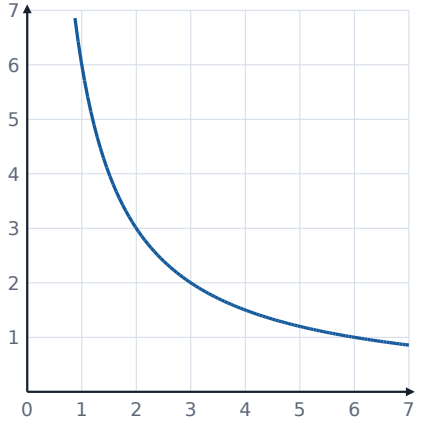
FORWARD 2

Estimate average rates of change from a rational-function graph • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows f(x) = 6 ÷ x.

Estimate the average rate of change from x = 1 to x = 3.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q1/5

1490

QUESTION ID

S79-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form. (2 − 6) ÷ 2.

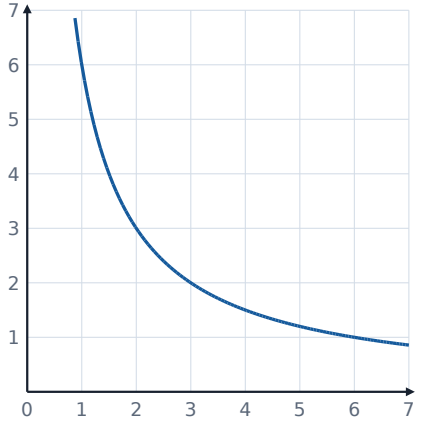
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows f(x) = 6 ÷ x.

Estimate the average rate of change from x = 3 to x = 6.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q2/5

1491

QUESTION ID

S79-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/3

GRADING NOTE Any equivalent form (about −0.33). (1 − 2) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows f(x) = 6 ÷ x.

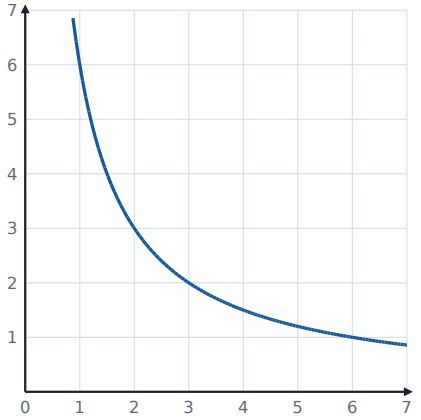
What happens to its average rate of change on intervals farther to the right?

A. It gets closer to 0.

B. It becomes positive.

C. It stays −2.

D. It gets more negative.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q3/5

1492

QUESTION ID

S79-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. It gets closer to 0.

GRADING NOTE The graph flattens toward its horizontal asymptote.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q3/5

True or false?

The graph of $y = 6 \div x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q4/5

1493

QUESTION ID

S79-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $6 \div x$ approaches 0 as x grows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 6 \div x$ passes through (0, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q5/5

1494

QUESTION ID
S79-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 6 ÷ 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph linear functions and show their intercepts

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Find slope and write y = mx + b	**8.EE.B.6**
BACKWARD 2	Write linear equations in two variables from a situation	**HSA-CED.A.2**
FORWARD 1	Find a point that partitions a segment in a given ratio	**HSG-GPE.B.6**
FORWARD 2	Find and verify inverses of linear functions	**HSF-BF.B.4.b**

SET 41 • HSF-IF.C.7.a

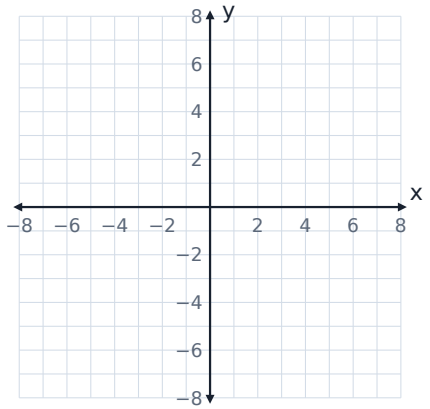
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = -\frac{2}{3}x + 4$ on the coordinate plane.

Give the x-intercept and the y-intercept.



Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q1/5

1497

QUESTION ID

S41-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line through (0, 4), (3, 2), and (6, 0)

Written: x-intercept (6, 0); y-intercept (0, 4)

GRADING NOTE Form required: both intercepts. Both the graph and the intercepts are graded. $-\frac{2}{3}x + 4 = 0$ gives $x = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the x- and y-intercepts of 3x − 5y = 15.

Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q2/5

1498

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 0) and (0, −3)

GRADING NOTE Form required: both intercepts. Set y = 0, then x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through (−2, 7) and (4, −5) in slope-intercept form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q3/5

1499

QUESTION ID

S41-M-Q3

TYPE

Short answer

CORRECT ANSWER

y = −2x + 3

GRADING NOTE Form required: slope-intercept form, y = mx + b. The slope is −12 ÷ 6 = −2, and 7 = −2(−2) + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q3/5

Which describes the graph of y = −3?

A. A line with slope 1 through (−3, −3)

B. A vertical line through (−3, 0)

C. A line with slope −3 through the origin

D. A horizontal line through (0, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q4/5

1500

QUESTION ID

S41-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A horizontal line through (0, −3)

GRADING NOTE y is −3 for every x, and the slope is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q4/5

True or false?

The graph of $2x + y = 6$ has x-intercept 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q5/5

1501

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At $y = 0$, $2x = 6$, so the x-intercept is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 1

Find slope and write y = mx + b • Grade 8 • Expressions & Equations • 8.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (3, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q1/5

1503

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 6 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line has slope 2 and y-intercept −1.

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q2/5

1504

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x − 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q2/5

What is the y-intercept of y = 5x − 4?

A. $\frac{4}{5}$

B. 4

C. 5

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q3/5

1505

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −4

GRADING NOTE y = mx + b with b = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Any two distinct points on a nonvertical line give the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q4/5

1506

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similar slope triangles have equal ratios. (A vertical line has no slope.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q4/5

True or false?

The line y = 4x passes through (1, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q5/5

1507

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4(1) = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 2

Write linear equations in two variables from a situation • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges $25 to join plus $15 per month.

Write an equation for the cost C after m months.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q1/5

1509

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

C = 25 + 15m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A candle 20 cm tall burns 2 cm per hour.

Write an equation for its height h after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q2/5

1510

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

h = 20 − 2t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Ana earns $8 per hour plus $5 in tips.

Which equation gives her earnings E for h hours?

A. E = 8h + 5

B. E = 5h + 8

C. E = 13h

D. E = 8(h + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q3/5

1511

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. E = 8h + 5

GRADING NOTE 8 per hour, plus 5 once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q3/5

True or false?

d = 60t gives the distance traveled at 60 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q4/5

1512

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distance = rate × time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q4/5

True or false?

y = 100 − 4x describes a quantity that grows by 4 each step.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q5/5

1513

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 4 each step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a

FORWARD 1

Find a point that partitions a segment in a given ratio • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (−2, 3) to (6, 7).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q1/5

1515

QUESTION ID
S41-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(2, 5)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q1**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the point $\frac{1}{4}$ of the way from A(0, 0) to B(8, 12).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q2/5

1516

QUESTION ID
S41-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(2, 3)

GRADING NOTE Any equivalent form. $\frac{1}{4}$ of (8, 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q2/5

Point P divides the segment from A(1, 2) to B(7, 11) in the ratio 1 : 2 (AP : PB).

Find P.

A. (5, 8)

B. (3, 5)

C. (4, 6.5)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q3/5

1517

QUESTION ID
S41-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. (3, 5)

GRADING NOTE P is $\frac{1}{3}$ of the way: (1 + 2, 2 + 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q3/5

True or false?

The midpoint of (0, 0) and (10, 4) is (5, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q4/5

1518

QUESTION ID
S41-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Average the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q4/5

True or false?

The point $\frac{1}{3}$ of the way from (0, 0) to (9, 6) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S41-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q5/5

1519

QUESTION ID
S41-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is (3, 2); (6, 4) is $\frac{2}{3}$ of the way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S41-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSF-IF.C.7.a

FORWARD 2

Find and verify inverses of linear functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

Find the inverse of f(x) = 3x − 6.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q1/5

1521

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 6) ÷ 3

GRADING NOTE Any equivalent form, such as $\frac{1}{3}x + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q1/5

Show that f(x) = 2x + 1 and g(x) = (x − 1) ÷ 2 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q2/5

1522

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f(g(x)) = 2 · (x − 1) ÷ 2 + 1 = x, and g(f(x)) = (2x + 1 − 1) ÷ 2 = x.

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the inverse of f(x) = x ÷ 4 + 1?

A. $f^{-1}(x) = 4x + 1$

B. $f^{-1}(x) = 4(x + 1)$

C. $f^{-1}(x) = 4x - 4$

D. $f^{-1}(x) = x \div 4 - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q3/5

1523

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $f^{-1}(x) = 4x - 4$

GRADING NOTE Solve x = y ÷ 4 + 1 for y: y = 4(x − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q3/5

True or false?

If f(g(x)) = x and g(f(x)) = x for every x, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q4/5

1524

QUESTION ID
S41-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = x + 5$ is $f^{-1}(x) = x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q5/5

1525

QUESTION ID
S41-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The inverse is $x - 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q5/5

SET 42 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph quadratic functions and show intercepts, maxima, and minima

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Factor to find the zeros of a quadratic	**HSA-SSE.B.3.a**
BACKWARD 2	Interpret key features of a function	**HSF-IF.B.4**
FORWARD 1	Relate a parabola's equation to its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use factors to describe the graph of a polynomial	**HSF-IF.C.7.c**

SET 42 • HSF-IF.C.7.a

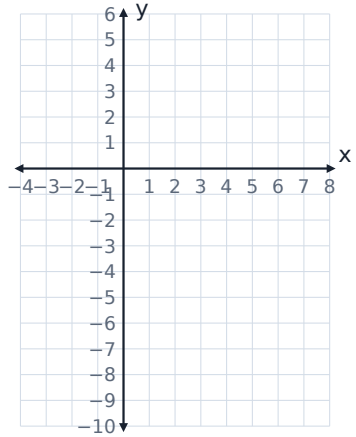
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = x^2 - 4x - 5$.

Give the vertex, the x-intercepts, and the y-intercept.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q1/5

1528

QUESTION ID

S42-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A parabola opening up with vertex (2, −9), through (−1, 0), (0, −5), (4, −5), and (5, 0)

Written: Vertex (2, −9); x-intercepts (−1, 0) and (5, 0); y-intercept (0, −5)

GRADING NOTE Form required: all four points. Both the graph and the points are graded. $f(x) = (x - 5)(x + 1) = (x - 2)^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q1/5

Find the vertex and the axis of symmetry of $y = 2x^2 + 8x + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q2/5

1529

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

Vertex (−2, −5); axis x = −2

GRADING NOTE Form required: both. $x = -b \div (2a) = -8 \div 4 = -2$, and $y = 8 - 16 + 3 = -5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Does $y = -3x^2 + 6x - 1$ have a maximum or a minimum? Find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q3/5

1530

QUESTION ID

S42-M-Q3

TYPE

Short answer

CORRECT ANSWER

A maximum of 2, at x = 1

GRADING NOTE Form required: maximum or minimum, its value, and where. $a = -3 < 0$, so the parabola opens down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q3/5

What are the x-intercepts of $y = (x + 3)(x - 1)$?

A. (3, 0) and (−1, 0)

B. (−3, 0) and (1, 0)

C. (−3, 0) and (−1, 0)

D. (0, −3) only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q4/5

1531

QUESTION ID

S42-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (−3, 0) and (1, 0)

GRADING NOTE $y = 0$ when $x + 3 = 0$ or $x - 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q4/5

True or false?

The graph of $y = x^2 + 4$ has no x-intercepts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q5/5

1532

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 + 4 \geq 4$ for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 1

Factor to find the zeros of a quadratic • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - 4x - 5$, and give its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q1/5

1534

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 5)(x + 1); zeros 5 and −1

GRADING NOTE Form required: the factored form and both zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q1/5

Find the zeros of $2x^2 - 8$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q2/5

1535

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2 and −2

GRADING NOTE Form required: both zeros. $2(x - 2)(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q2/5

What are the zeros of $x^2 + 6x + 8$?

A. 2 and −4

B. −2 and −4

C. 2 and 4

D. −2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q3/5

1536

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −2 and −4

GRADING NOTE (x + 2)(x + 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q3/5

True or false?

The zeros of $(x - 7)(x + 2)$ are 7 and -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q4/5

1537

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each factor is 0 at one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q4/5

True or false?

The zeros of $x^2 - 9x$ are 0 and −9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q5/5

1538

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x(x − 9): the zeros are 0 and 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 2

Interpret key features of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A table has x = −2, −1, 0, 1, 2 and y = 5, 2, 1, 2, 5.

Describe its symmetry and its minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q1/5

1540

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Symmetric about x = 0; minimum 1 at x = 0.

GRADING NOTE Form required: both features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q1/5

A profit function P(x) is positive for $10 < x < 50$, where x is the number of items sold.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q2/5

1541

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The company makes a profit when it sells between 10 and 50 items.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q2/5

The highest point of a graph is (3, 40).

Which statement describes it?

A. A maximum of 40 at x = 3

B. A maximum of 3 at x = 40

C. A minimum of 3 at x = 40

D. A y-intercept of 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q3/5

1542

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A maximum of 40 at x = 3

GRADING NOTE The maximum value is the output, 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q3/5

True or false?

A function is increasing on an interval when its outputs get larger as its inputs get larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q4/5

1543

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q4/5

True or false?

The x-intercepts of a graph are where the output is greatest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q5/5

1544

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The output is 0 at an x-intercept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 1

Relate a parabola's equation to its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = \frac{1}{8}x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q1/5

1546

QUESTION ID
S42-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Focus (0, 2); directrix y = −2

GRADING NOTE Form required: both. $x^2 = 8y = 4py$ with $p = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive the equation of the parabola with focus $(0, -1)$ and directrix $y = 1$ from the distance definition.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q2/5

1547

QUESTION ID
S42-F1-Q2

TYPE
Answer with required work

CORRECT ANSWER
$x^2 = -4y$

GRADING NOTE REQUIRED WORK: $\sqrt{(x^2 + (y + 1)^2)} = |y - 1|$ written, both sides squared, and simplified: $x^2 + 2y = -2y$, so $x^2 = -4y$.

Form required: the derivation and the equation (any equivalent form, such as $y = -\frac{1}{4}x^2$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A parabola has focus (0, 3) and directrix y = −3.

Where is its vertex?

A. (0, 0)

B. (0, 3)

C. (0, −3)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q3/5

1548

QUESTION ID
S42-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. (0, 0)

GRADING NOTE Halfway between the focus and the directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q3/5

True or false?

The parabola $x^2 = 12y$ opens upward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q4/5

1549

QUESTION ID
S42-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $p = 3 > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q4/5

True or false?

The parabola with focus (0, 1) and directrix $y = -1$ is $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q5/5

1550

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = 4y$, so $y = \frac{1}{4}x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 2

Use factors to describe the graph of a polynomial • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

$f(x) = (x + 1)(x - 2)(x - 4)$.

Find the zeros and the y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q1/5

1552

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Zeros −1, 2, and 4; y-intercept 8

GRADING NOTE Form required: all three zeros and the y-intercept. f(0) = (1)(−2)(−4) = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

At most how many turning points can the graph of a degree-3 polynomial have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q2/5

1553

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. At most n − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q2/5

$f(x) = (x - 3)^2(x + 1)$.

What does the graph do at $x = 3$?

A. It touches the x-axis and turns around.

B. It is undefined.

C. It has a vertical asymptote.

D. It crosses the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q3/5

1554

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. It touches the x-axis and turns around.

GRADING NOTE The factor $(x - 3)$ is squared, so f does not change sign at 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^3 - 8$ has exactly one real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q4/5

1555

QUESTION ID

S42-F2-Q4

TYPE

True or false

CORRECT ANSWER

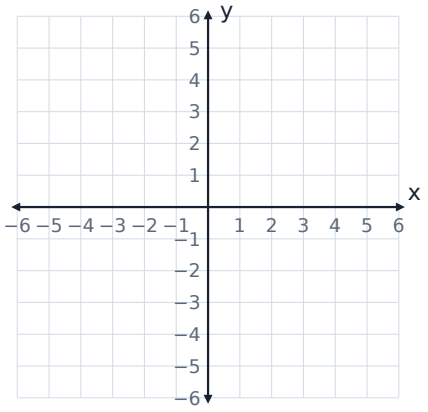
A. True

GRADING NOTE $x^3 = 8$ gives only $x = 2$ among the real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q4/5

Graph $f(x) = (x + 2)(x - 1)^2$. Mark the zeros and the y-intercept, and show what the graph does at each zero. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q5/5

1556

QUESTION ID

S42-F2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −2, touching it at 1 and turning back up, through (0, 2), falling to the left and rising to the right

GRADING NOTE Any correct graph. f(0) = (2)(1) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph absolute value functions and show their key features

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Find the domain and range of a function	**HSF-IF.A.1**
BACKWARD 2	Understand absolute value as distance from 0	**6.NS.C.7.c**
FORWARD 1	Solve radical equations and check for extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Read values of inverses of power and root functions	**HSF-BF.B.4.c**

SET 43 • HSF-IF.C.7.b

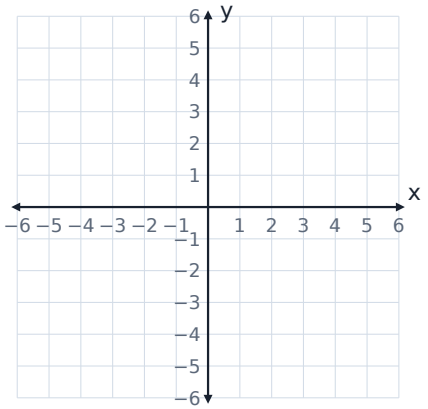
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

Graph f(x) = |x − 2| − 3.

Give the vertex and the x-intercepts.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q1/5

1559

QUESTION ID

S43-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A V-shape opening up with vertex (2, −3), through (−1, 0), (0, −1), (4, −1), and (5, 0)

Written: Vertex (2, −3); x-intercepts (−1, 0) and (5, 0)

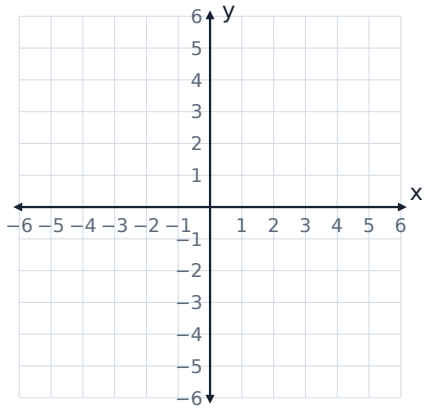
GRADING NOTE Form required: the vertex and both intercepts. Both the graph and the answer are graded. |x − 2| = 3 gives x = 5 or x = −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph g(x) = −2|x + 1| + 4, and label its vertex and its x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q2/5

1560

QUESTION ID

S43-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A V-shape opening down with vertex (−1, 4), slopes 2 and −2, crossing the x-axis at (−3, 0) and (1, 0)

GRADING NOTE Any correct graph. |x + 1| = 2 gives x = 1 or x = −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe how the graph of $y = |x + 3| + 1$ is related to the graph of $y = |x|$, and give its range.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q3/5

1561

QUESTION ID
S43-M-Q3

TYPE
Short answer

CORRECT ANSWER
It is the graph of $y = |x|$ shifted 3 units left and 1 unit up (vertex $(-3, 1)$); range $y \geq 1$.

GRADING NOTE Form required: the shifts and the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q3/5

Which function's graph is a V opening downward with vertex (2, 5)?

A. $y = |x - 2| + 5$

B. $y = -|x + 2| + 5$

C. $y = -|x - 5| + 2$

D. $y = -|x - 2| + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q4/5

1562

QUESTION ID

S43-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = -|x - 2| + 5$

GRADING NOTE $y = -|x - h| + k$ opens downward with vertex $(h, k) = (2, 5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q4/5

True or false?

The graph of $y = 3|x|$ is narrower than the graph of $y = |x|$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q5/5

1563

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each output is tripled, so the sides are steeper (slopes 3 and −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 1

Find the domain and range of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Over the real numbers, what is the domain of $f(x) = \sqrt{(x - 5)}$?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q1/5

1565

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 5$

GRADING NOTE Form required: an inequality (or interval).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of f(x) = |x|?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q2/5

1566

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y ≥ 0

GRADING NOTE Form required: an inequality (or interval).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q2/5

Which value is NOT in the domain of $g(x) = \sqrt{(6 - x)}$?

A. 7

B. −3

C. 0

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q3/5

1567

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7

GRADING NOTE $6 - 7 = -1$ is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

Every real number is in the domain of $f(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q4/5

1568

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every real number has a real cube root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

The range of $f(x) = x^2$ is all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q5/5

1569

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The range is $y \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 2

Understand absolute value as distance from 0 • Grade 6 • The Number System • 6.NS.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

What is |−9|?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q1/5

1571

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. −9 is 9 units from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is farther from 0, −7 or 5? Explain with absolute value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q2/5

1572

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−7, because |−7| = 7 and |5| = 5.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q2/5

Which value equals |3 − 8|?

A. 5

B. −5

C. 11

D. −11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q3/5

1573

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE |−5| = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q3/5

True or false?

$|-4| = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q4/5

1574

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −4 is 4 units from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$|0| = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q5/5

1575

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0 is 0 units from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 1

Solve radical equations and check for extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve √(x − 2) = 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q1/5

1577

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form. x − 2 = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt{(x + 6)} = x$. Name any extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q2/5

1578

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3; x = −2 is extraneous.

GRADING NOTE Form required: the solution and the extraneous value. $x^2 - x - 6 = 0$ gives 3 and −2, but $\sqrt{4} = 2 \neq -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt[3]{(x + 1)} = 2$.

A. x = 7

B. x = 3

C. x = 9

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q3/5

1579

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 7

GRADING NOTE x + 1 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q3/5

True or false?

$\sqrt{x} + 3 = 0$ has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q4/5

1580

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{x} \geq 0$, so $\sqrt{x} + 3 \geq 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 16$ is a solution of $\sqrt{x} = -4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q5/5

1581

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{16} = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 2

Read values of inverses of power and root functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

$g(x) = x^3$.

Find $g^{-1}(27)$.

$g^{-1}(27) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q1/5

1583

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (4, 2) is on the graph of $f(x) = \sqrt{x}$.

Which point must be on the graph of f^{-1}?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q2/5

1584

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 4)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q2/5

$f(x) = x^2$ with domain $x \geq 0$.

What is $f^{-1}(49)$?

A. -7

B. 24.5

C. 2,401

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q3/5

1585

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE $f(7) = 49$, and 7 is in the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q3/5

True or false?

The inverse of $f(x) = x^3$ is $f^{-1}(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q4/5

1586

QUESTION ID
S43-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Cubing and taking the cube root undo each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q4/5

True or false?

f is a one-to-one function. If (3, 10) is on the graph of f, then (3, 10) must be on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q5/5

1587

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (10, 3) must be on the graph of f^{-1}, but (3, 10) need not be: for f(x) = x + 7, $f^{-1}(3) = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root and cube root functions

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Find domains and ranges of root functions	**HSF-IF.A.1**
BACKWARD 2	Evaluate square roots and cube roots	**8.EE.A.2**
FORWARD 1	Solve radical equations and check the solutions	**HSA-REI.A.2**
FORWARD 2	Find inverses of square root and cube root functions	**HSF-BF.B.4.c**

SET 74 • HSF-IF.C.7.b

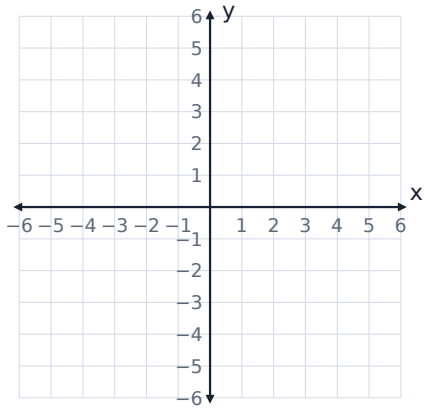
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

Graph f(x) = √(x + 4).

Give its domain and range.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q1/5

1590

QUESTION ID

S74-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A curve starting at (−4, 0) and rising to the right through (−3, 1), (0, 2), and (5, 3)

Written: Domain x ≥ −4; range y ≥ 0

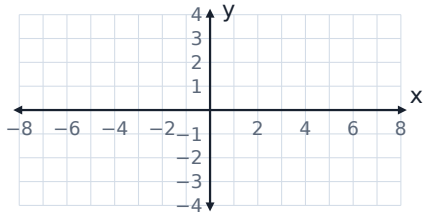
GRADING NOTE Form required: the domain and the range. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph $g(x) = \sqrt[3]{x}$ for $-8 \le x \le 8$. Plot the points where x is a perfect cube, then draw the curve.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q2/5

1591

QUESTION ID

S74-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A curve through (−8, −2), (−1, −1), (0, 0), (1, 1), and (8, 2), increasing everywhere and flattest away from the origin

GRADING NOTE Any correct graph through these five points.

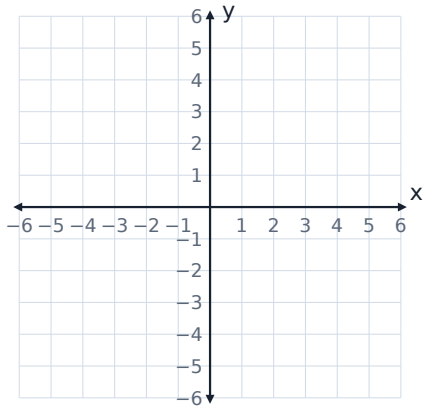
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph h(x) = −√x + 3.

Give its starting point and its range.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q3/5

1592

QUESTION ID

S74-M-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A curve starting at (0, 3) and falling to the right through (1, 2) and (4, 1)

Written: Starting point (0, 3); range y ≤ 3

GRADING NOTE Form required: the starting point and the range. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q3/5

Which describes the graph of $y = \sqrt[3]{x}$?

A. It starts at (0, 0) and goes only to the right.

B. It has the horizontal asymptote y = 0.

C. It passes through (−8, −2), (0, 0), and (8, 2).

D. It is V-shaped with vertex (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q4/5

1593

QUESTION ID

S74-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. It passes through (−8, −2), (0, 0), and (8, 2).

GRADING NOTE Cube roots of negative numbers are negative, so the graph extends in both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of $y = \sqrt{(x - 1)}$ starts at $(-1, 0)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q5/5

1594

QUESTION ID

S74-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x - 1)}$ is defined for $x \geq 1$, so the graph starts at $(1, 0)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q5/5

SET 74 • HSF-IF.C.7.b

BACKWARD 1

Find domains and ranges of root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Over the real numbers, what is the domain of $g(x) = \sqrt{(2x - 6)}$?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q1/5

1596

QUESTION ID

S74-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 3$

GRADING NOTE Form required: an inequality (or interval). $2x - 6 \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of h(x) = √x + 2?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q2/5

1597

QUESTION ID

S74-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y ≥ 2

GRADING NOTE Form required: an inequality (or interval).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q2/5

What is the domain of $y = \sqrt[3]{(x - 5)}$?

A. All real numbers

B. $x \leq 5$

C. $x > 5$

D. $x \geq 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q3/5

1598

QUESTION ID

S74-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. All real numbers

GRADING NOTE Every real number has a real cube root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The range of $y = -\sqrt{x}$ is $y \leq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q4/5

1599

QUESTION ID

S74-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{x} \geq 0$, so $-\sqrt{x} \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q4/5

1599

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = -1$ is in the domain of $y = \sqrt{x}$ over the real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q5/5

1600

QUESTION ID

S74-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(-1)}$ is not a real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q5/5

1600